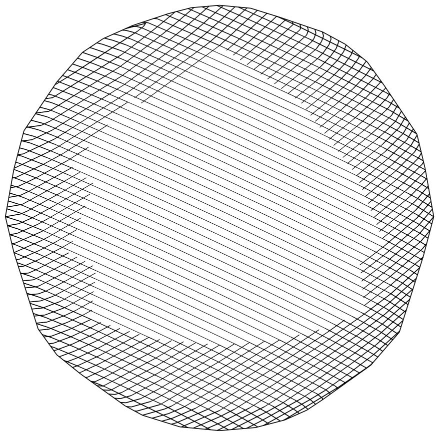
PLATE 01

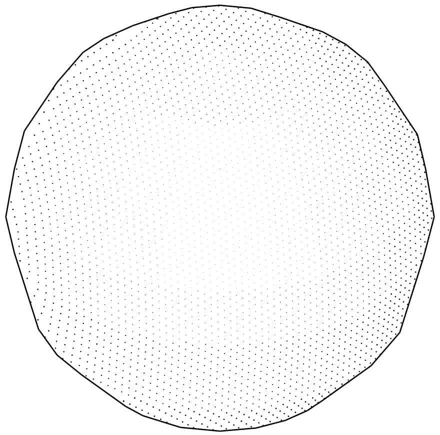
The receiver gives the figure its face

One geometric carrier; line engraving, discrete stippling and coherent phase response are different readings.

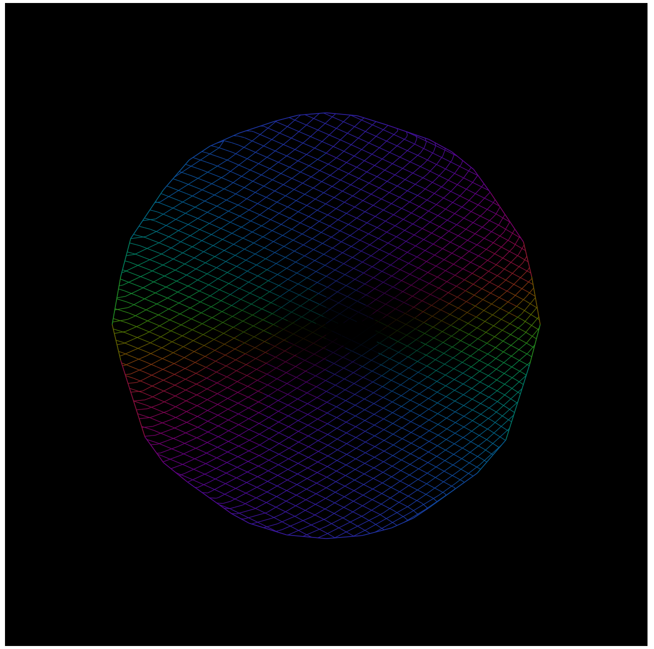
Field contours and foreshortening



Intersections of the same level families



The complex current, read on black



$(K, Z, U) \to R \to (\text{visible fibre}, \chi_0, \chi_1, P) \to \text{engraved face}$

The geometry is upstream of the texture.

Exact rational stations and triangular incidence define the finite carrier. Hatches are level traces of supplied receiver fields. The dot pattern uses intersections of those traces; no random seed places structure. The visible mesh is a finite realization of the rational sphere chart.

The image keeps a source behind it.

The packet retains complex vertices, currents, face addresses, the receiver and display error. Opaque-surface visibility is decided before paint. A black or unmarked region can be a zero response, an occlusion or a receiver omission; these remain different source records.

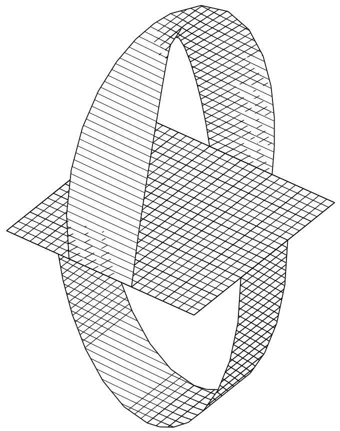
Method recovered. Fiziko varies stroke width, generator families and visibility to engrave curved objects; vintage-latex separates contours, texture and labels. These are original Typst packages using the existing Holonic receiver laws. White paper is fixed; phase frames are pure black. Sources: Fiziko, vintage-latex.

PLATE 02

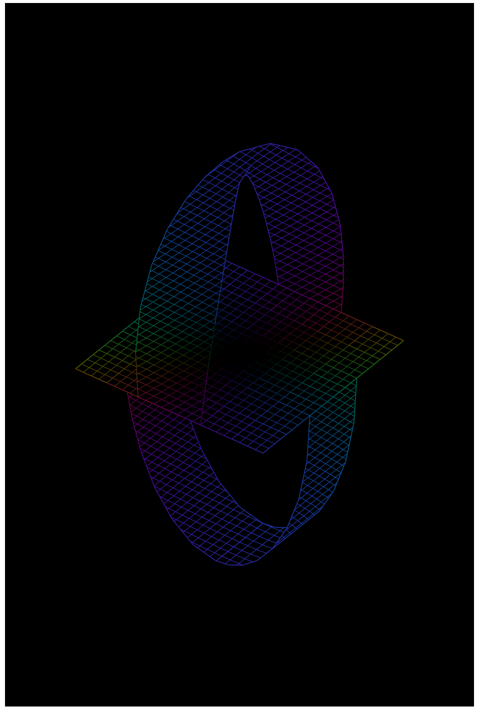
Curvature, folding and orientation have different sources

Curved rational ribbon charts preserve their attachments; the receiver carries their local current into an image.

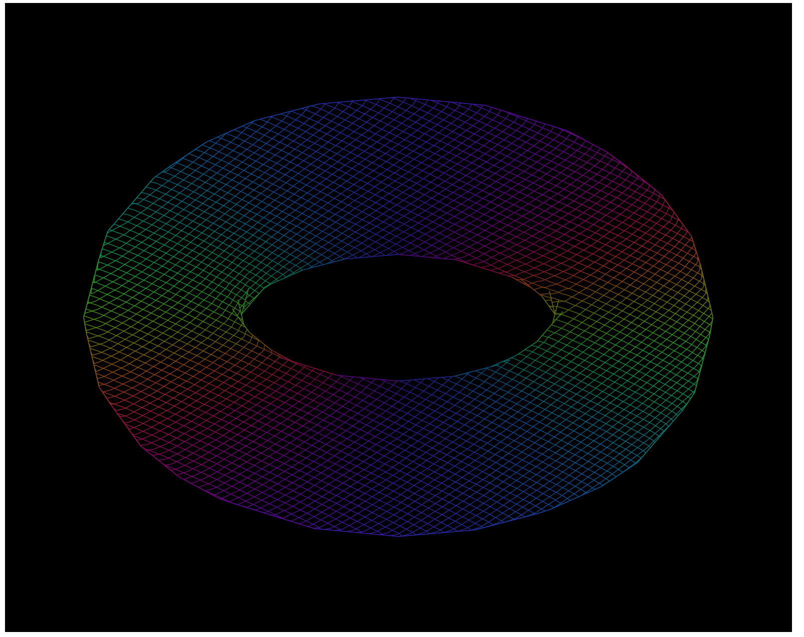
Möbius shorts · monochrome



The same packet geometry · phase



A closed orientable carrier



$$\chi(\text{torus}) = 0, \quad \chi(\text{shorts}) = -1$$

The torus is orientable. Shorts have one boundary and an orientation-reversing loop. Their appearance under projection proves neither fact; the incidence witness does.

$$\varepsilon_g = -h_{fe}h_{ge}\varepsilon_f, \quad \prod_{\gamma}(-h_{fe}h_{ge}) = -1$$

Exact finite construction. The ribbon uses a rational transverse frame whose squared length is $1 - \frac{3\sin^4 t}{4}$. Its width therefore varies along the curved chart. The earlier unit-normal ribbon and its chosen $\frac{1}{12}$ shell offset remain in the companion atlas.

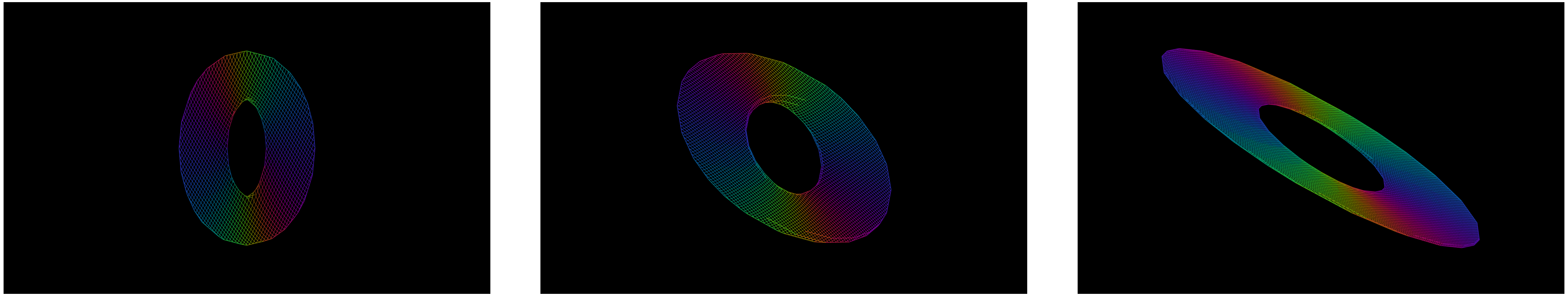
Receiver distinction. A polyhedral source has curvature concentrated at its edges. A projection may flatten a curved source. Intrinsic nonorientability is a separate failure of global orientation, while local transport and receiver-relative orientation remain available.

PLATE 03

A complete complex current drives the motion

An exact holomorphic Euler–NS source on C³; no Fourier-reality condition discards the imaginary field.

$\boldsymbol{\tau = 0}$ $\boldsymbol{\tau = \frac{1}{2}}$ $\boldsymbol{\tau = 1}$



$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & \mathrm{i} \\ 1 & \mathrm{i} & 0 \end{pmatrix}, \quad A^3 = 0, \quad U(Z) = AZ, \quad P(Z) = -\frac{1}{2}(Z_1 + \mathrm{i}Z_2)^2$$

$$F_\tau = I + \tau A + \frac{\tau^2}{2}A^2, \quad F_{-\tau}F_\tau = I, \quad \det_{\mathbb{C}} F_\tau = 1$$

$$(U \cdot \nabla)U = A^2 Z = -\nabla P, \quad \Delta U = 0, \quad \nabla \cdot U = 0$$

The two fields interact.

$$a_t + (a \cdot \nabla)a - (b \cdot \nabla)b = \nu\Delta a - \nabla p$$

$$b_t + (a \cdot \nabla)b + (b \cdot \nabla)a = \nu\Delta b - \nabla q$$

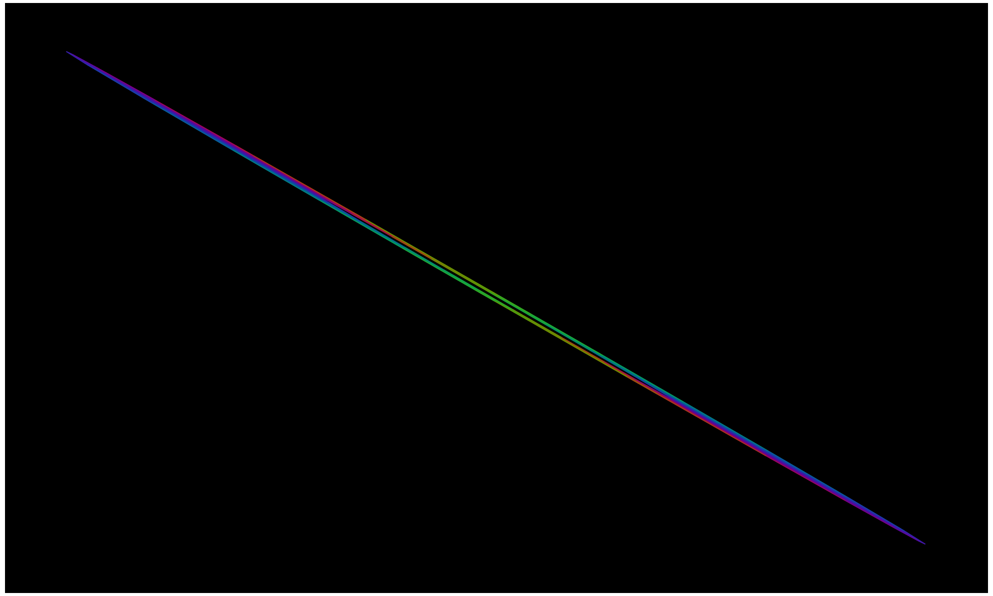
Proved-derived. $U = a + \mathrm{i}b$, $P = p + \mathrm{i}q$; advection is complex-bilinear. The holomorphic differential is $\Delta = \sum_j \partial^2_{Z_j}$. This polynomial field satisfies the displayed equation for every declared viscosity. The receiving image is a real projection of the full complex flow; the camera and drawing scale are held fixed.

PLATE 04

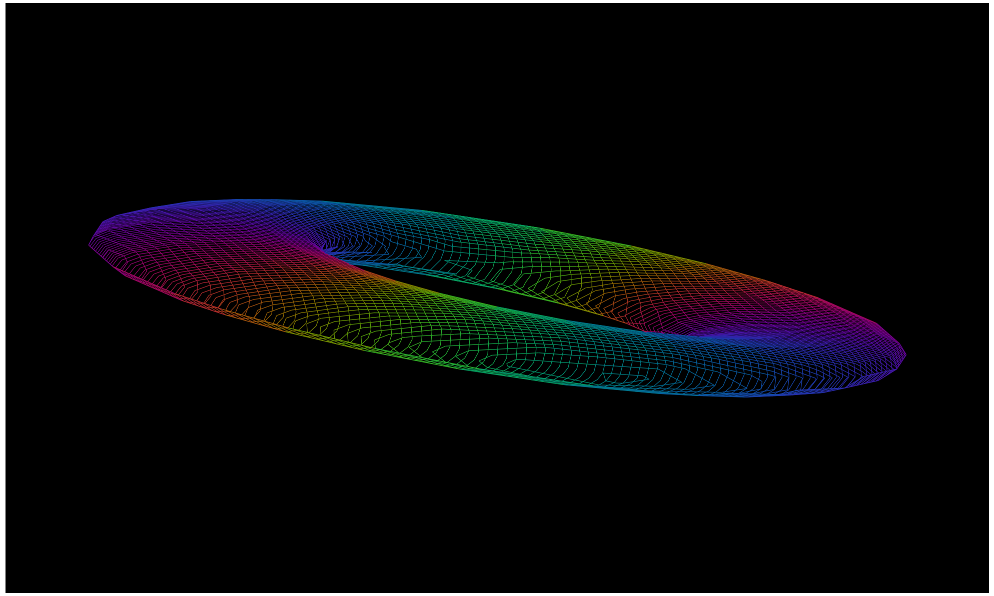
A singular image can have a regular source

Keep the complex fibre when a receiver loses directions. A second receiver can expose them again.

Real receiver · $\tau = \frac{7}{5}$



Changed receiver · same source and clock



$$R_0 F_\tau = \operatorname{Re} F_\tau, \quad \det_{\mathbb{R}}(R_0 F_\tau) = \left(1 - \frac{\tau^2}{2}\right)^2$$

$$R_{a,b}(Z) = a \operatorname{Re} Z + b \operatorname{Im} Z, \quad (a, b) = \left(\frac{3}{5}, \frac{4}{5}\right)$$

Proved-derived. At $\tau^2 = 2$ this real receiver has rank one, while F_τ is invertible. The picture uses rational $\tau = \frac{7}{5}$ near that exact singular cut. The source is not replaced by the nearly collapsed image; its complex coordinates and current are retained.

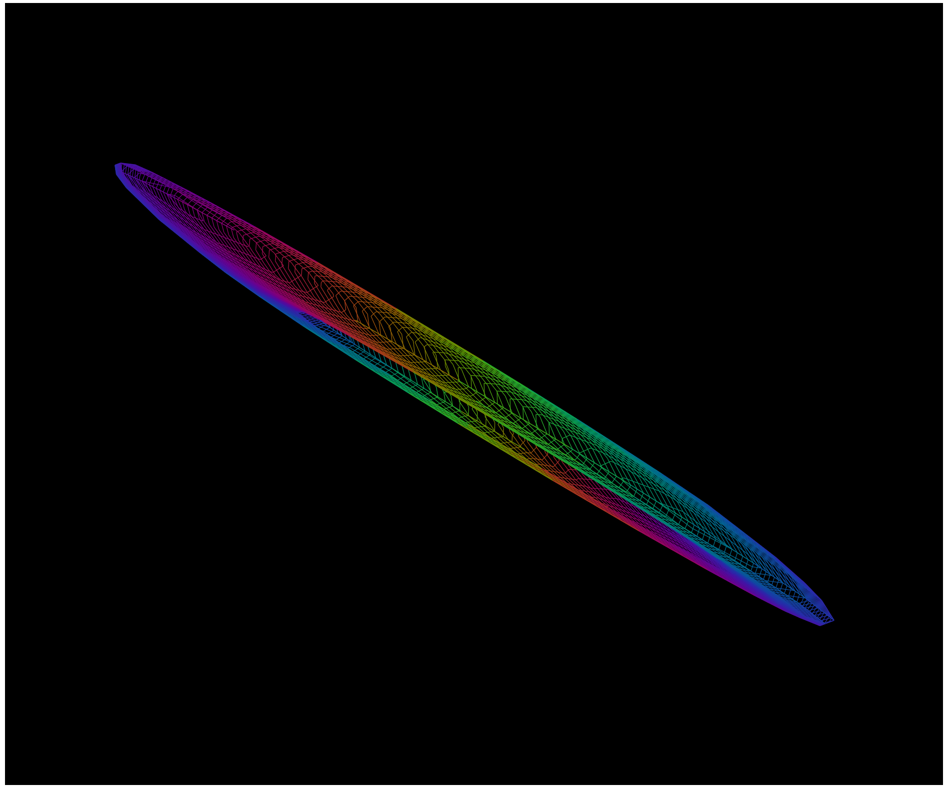
Actual folded source. On the real slice, $a = (x_3, 0, x_1)$ and $b = (0, x_3, x_2)$. Dropping b with the same declared real pressure drops $(b \cdot \nabla) b = (0, x_2, x_3)$. This is why the complex evolution cannot generally be continued from its present real face alone.

PLATE 05

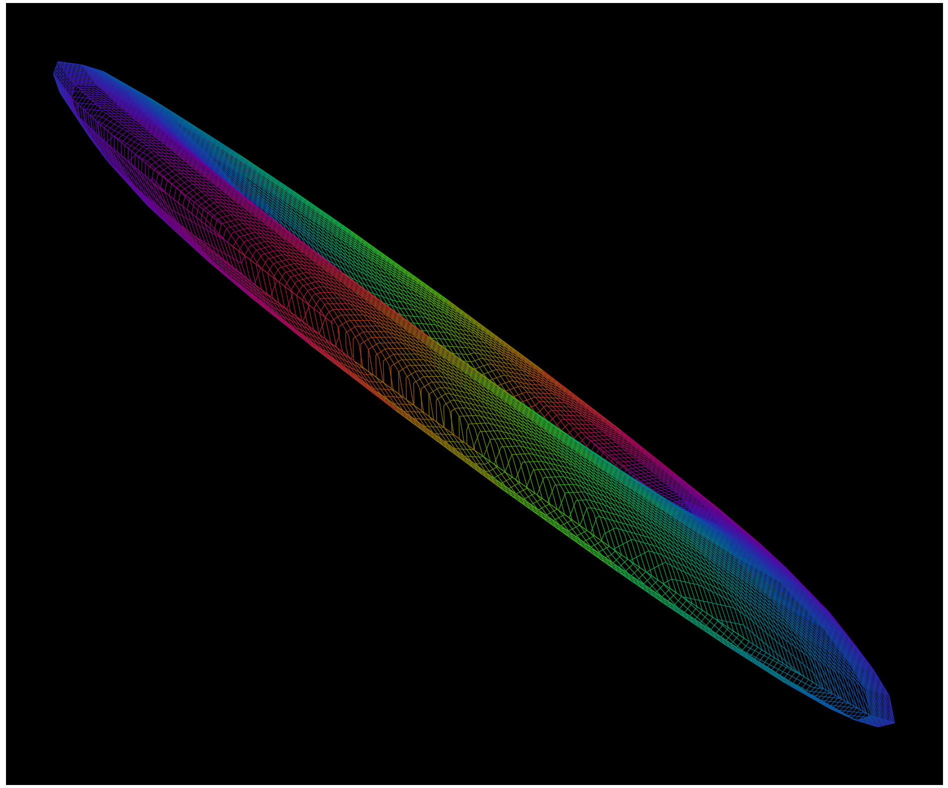
Current and curl have separate receivers

Add a declared rotational current to the same complex source family; read circulation through actual face boundaries.

$\boldsymbol{\lambda = \frac{1}{2}, \tau = \frac{7}{5}}$



$\boldsymbol{\lambda = 1, \tau = \frac{7}{5}}$



$$v = (1, \mathrm{i}, 0), \quad KZ = v \times Z, \quad A_\lambda = S + \lambda K, \quad U_\lambda = A_\lambda Z$$

$$A_\lambda^2 = (1 + \lambda^2) S^2, \quad A_\lambda^3 = 0, \quad \operatorname{curl} U_\lambda = 2\lambda v$$

$$j_{pq} = \frac{U_{\lambda(p)} + U_{\lambda(q)}}{2} \cdot (q - p), \quad \sum_{e \in \partial f} j_e = \operatorname{curl} U_\lambda \cdot \mathrm{area}_f$$

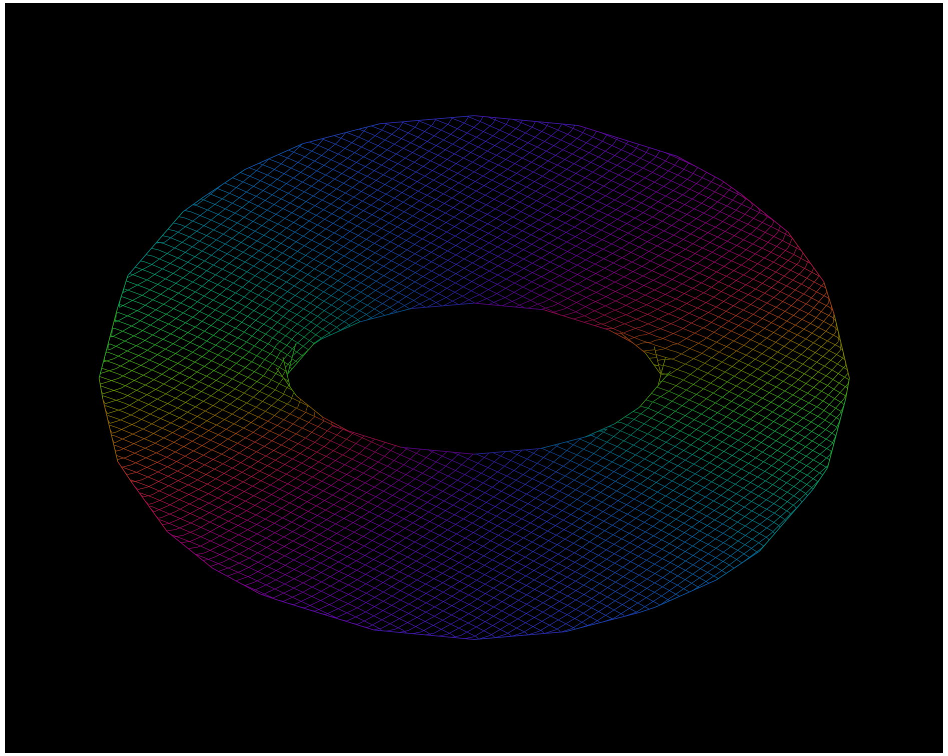
Exact complex Stokes instance. S is the matrix on plate 3; $P_\lambda = (1 + \lambda^2) P_0$. On the oriented triangle $(1,0,0) \to (0,1,0) \to (0,0,1)$, the boundary return is $\lambda(1 + \mathrm{i})$. Both channels are retained. The hatches/colors read the supplied current, while this independent edge integral reads its curl. The source geometry, rotational contribution and receiver are explicit; a triangle outline alone is no flux measurement.

PLATE 06

The cross-hatching carries a receiver differential

The line families can read current phase or distributed cross-entropy. Their source functions determine the marks.

Complex-current level families



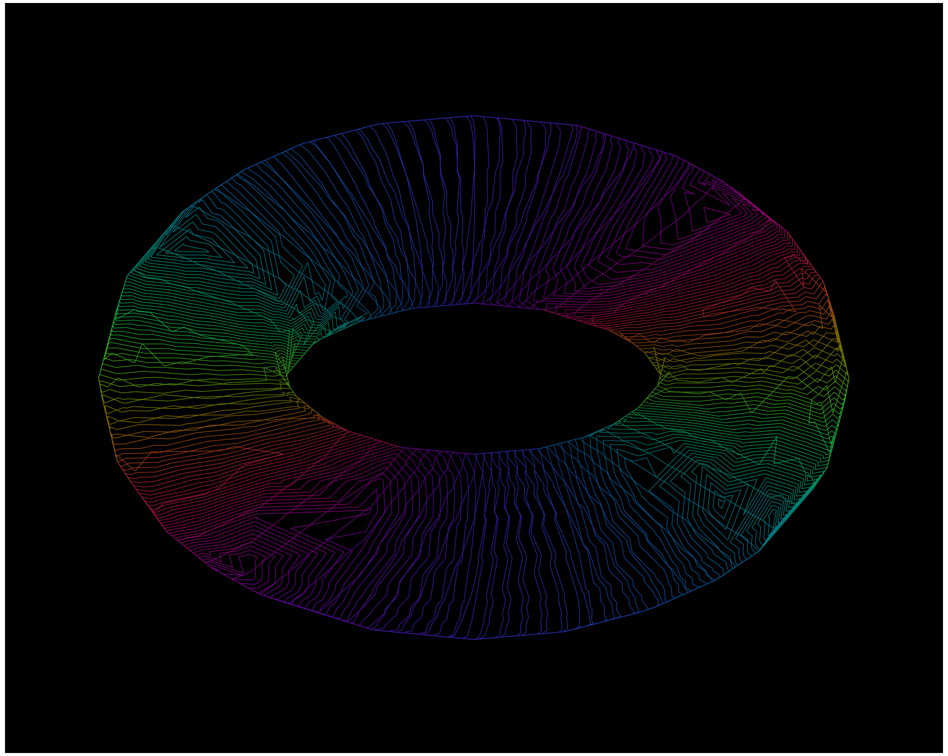
$$P = (a^2, b^2, (a+b)^2), \quad C = \frac{P}{\kappa + \sum_j P_j}, \quad \sum_j C_j + \frac{\kappa}{\kappa + \sum_j P_j} = 1$$

$$q_i = \frac{\frac{\kappa}{3} + P_i}{\kappa + \sum_j P_j}, \quad p_i = \frac{1}{3}, \quad h_i = -p_i \log q_i, \quad H = \sum_i h_i$$

$$dH = \sum_i (q_i - p_i)\,\mathrm{d}s_i \quad (q = \mathrm{softmax}(s))$$

Defined receiver. The right field uses h_0, h_1 as two level families, with the third contribution retained by the source law. Positive support makes the logs defined. Station logs carry rational enclosures; the rendered field is their declared affine triangle reading.

Distributed cross-entropy families



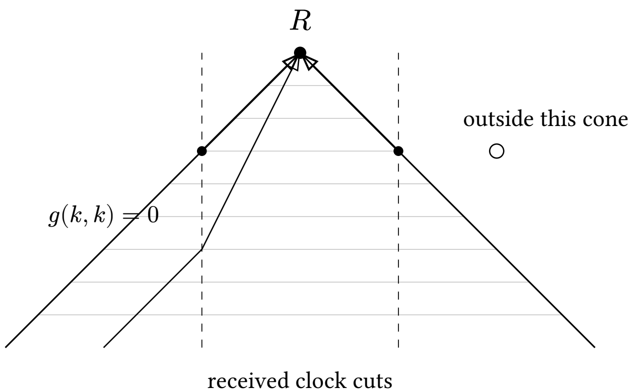
$$\int_\gamma d\chi = \chi(b) - \chi(a) = \Delta N + r_b - r_a, \quad |r_b - r_a| < \Delta$$

Proved-derived. Signed crossings of levels $k\Delta$ count the quantized potential flux. Endpoint remainders retain what finite line density omits. No hatch direction is assigned by semantic label, and a geometric slope is not silently renamed cross-entropy.

PLATE 07

The receiver is situated at an event

World-lines, cone membership, orientation and ordered frame changes belong to the reading.



$$g(u,u) = -1, \quad g(k,k) = 0, \quad \omega_R = -g(k,u)$$

Declared Lorentz receiver. A time-oriented metric, observer world-line and tetrad determine the received frequency and direction. Cone membership permits causal influence; actual incidence determines whether that influence occurs.

A curved metric changes the received frequency

$$f(r) = 1 - \frac{2}{r}, \quad g = \operatorname{diag}\big(-f, f^{-1}, r^2, r^2\big), \quad k = \big(f^{-1}, 1, 0, 0\big)$$

$$r_{\text{near}} = \frac{25}{8}, \quad r_{\text{far}} = \frac{50}{9}, \quad \omega_{\text{near}} = \frac{5}{3}, \quad \omega_{\text{far}} = \frac{5}{4}, \quad \frac{\omega_{\text{far}}}{\omega_{\text{near}}} = \frac{3}{4}$$

Exact Schwarzschild receiver instance. $G = c = M = 1$ at the equatorial slice; static tetrads are orthonormal, and the radial null ray has unit conserved Killing energy. Its two nontrivial geodesic equations cancel exactly. The frequency ratio follows from the metric and observer; no color is assigned to manufacture a redshift.

The gyroparallelogram retains the order

$$L_x = \begin{pmatrix} \frac{5}{4} & -\frac{3}{4} & 0 \\ -\frac{3}{4} & \frac{5}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$L_y = \begin{pmatrix} \frac{5}{3} & 0 & -\frac{4}{3} \\ 0 & 1 & 0 \\ -\frac{4}{3} & 0 & \frac{5}{3} \end{pmatrix}$$

$$L_y L_x \neq L_x L_y, \quad L^\top \eta L = \eta$$

$$B_{L_y L_x e_0} L_y L_x = \operatorname{diag}\left(1, \begin{pmatrix} \frac{35}{37} & -\frac{12}{37} \\ \frac{12}{37} & \frac{35}{37} \end{pmatrix}\right)$$

Exact rational return. The canonical return boost removes the final velocity; a spatial rotation remains. The 37 is produced by the two supplied boosts. This is a flat-spacetime frame instance; a curved GR receiver uses its supplied metric and connection.

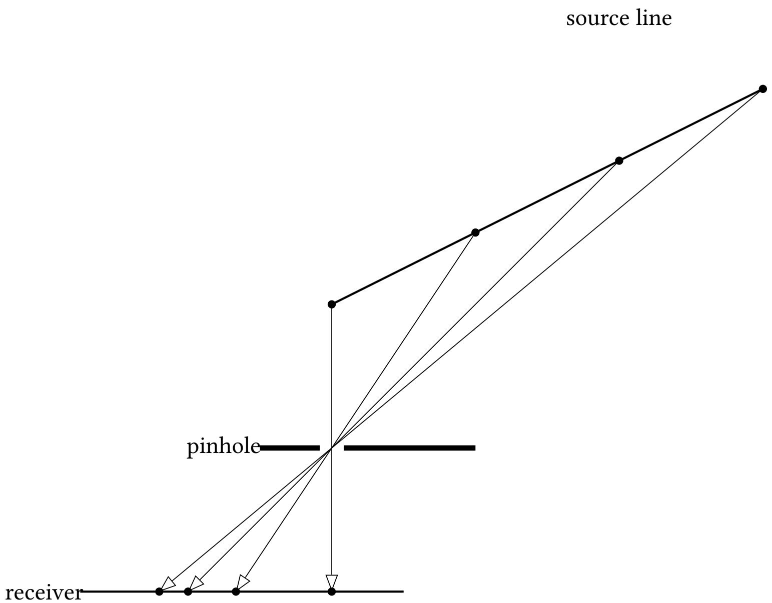
$$\nabla_a a - \nabla_b b, \quad \nabla_a b + \nabla_b a$$

Complex/GR interface. Complexifying a declared connection gives these two coupled advection terms. A holomorphic C^3 differential is a different chart from a Lorentzian space-time derivative. Their comparison retains the real restriction, tetrad, pressure/source return and any connection defect. No image supplies a stress-energy tensor or Einstein solution by appearance.

PLATE 08

A shadow retains a path behind its face

Projection, analytic continuation and a physical Feynman diagram have their own source maps.



$$f(t) = \frac{t}{1+\frac{t}{2}}, \quad \operatorname{cr}(0,1,2,3) = \operatorname{cr}\left(0, \frac{2}{3}, 1, \frac{6}{5}\right) = \frac{4}{3}$$

Exact projective instance. A noncollapsed line retains its cross-ratio. The pinhole denominator has a pole at $t = -2$; the source point remains defined. Projection does not preserve arbitrary lengths or reconstruct its fibre.

A hypergeometric face with a retained branch

$$F(z) = {}_2F_1(1,1;2;z) = -\frac{\log(1-z)}{z}$$

$$\begin{tikzcd}
(z,w) \arrow[r, "\text{around } 1"] \arrow[d] & (z,w+1) \arrow[d] \\
F_w \arrow[r] & F_w - 2\pi\frac{\mathrm{i}}{z}
\end{tikzcd}$$

$$F_w = F_0 - \frac{2\pi \mathrm{i} w}{z}, \quad w \in \mathbb{Z}$$

Derived continuation. A positive loop around 1 changes the logarithm by $2\pi\mathrm{i}$. The winding integer and source branch generate the family; the endpoint z alone does not identify it. Source: DLMF 15.4.1.

Recovered context. The August 20 “Shadows of Holonic Interactions” deposit refers to Tufte’s physical shadows of Feynman-diagram sculptures. Our formal owner proves polarity and conjugate-line identities. Hypergeometric monodromy supplies the separate analytic example above. Actual Feynman amplitudes require their propagators, couplings and integration/regularization data; a projected drawing does not supply them.

PLATE 09

The packet carries the construction

A reusable receiver package and an engraving package replace picture-specific paint rules.

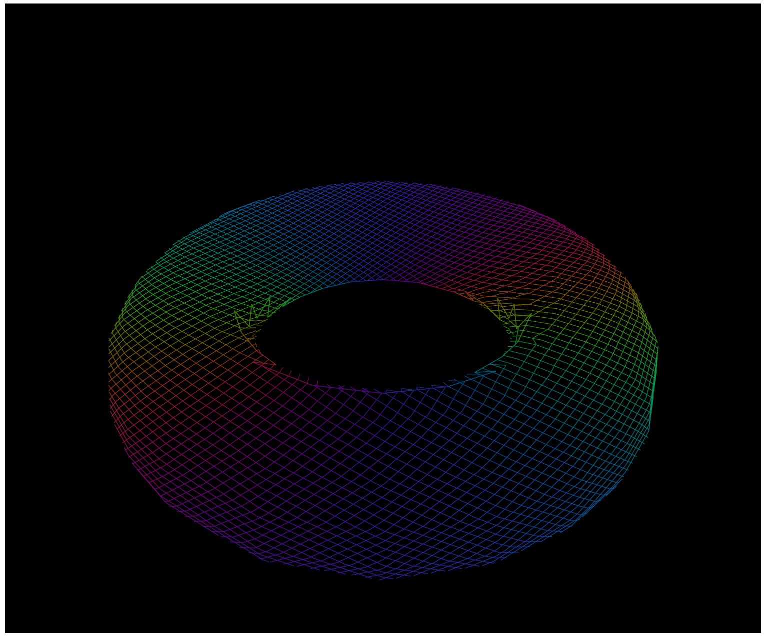
Complex vertices, current, incidence → Receiver, domain, phase, clock → Visible level traces and remainders → Typst vectors · SVG · raster face

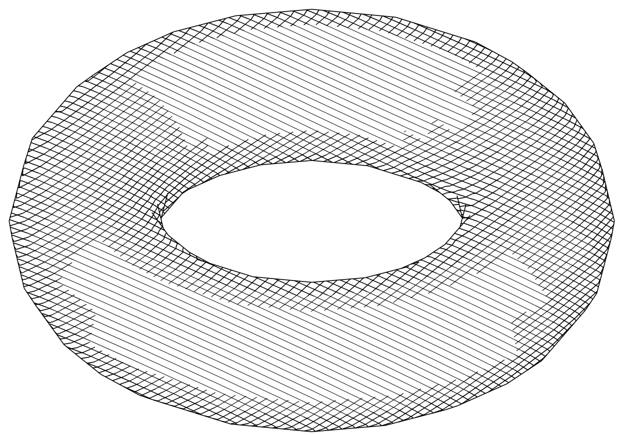
holonic-receiver

Exact rational companion: source projection, affine or reciprocal-depth visibility, level intersection, coherent primary response and enclosed entropy stations. Pure Typst helpers expose receiver maps and the softmax cross-entropy differential. The packet keeps source and omitted/degenerate populations.

holonic-engraving

CeTZ draws supplied marks. Line width, stippling and paint are explicit display choices; clipping and source identity are already in the packet. Phase colors are transferred without a contrast boost. The monochrome mode changes paint, not the supplied geometry.





$$qT_w = U_w q, \quad rT_w = \tilde{r} U_w q$$

$$C(A) = C(-A), \quad C(A+1) \neq C(-A+1) \quad (A = 1 + 2\mathrm{i})$$

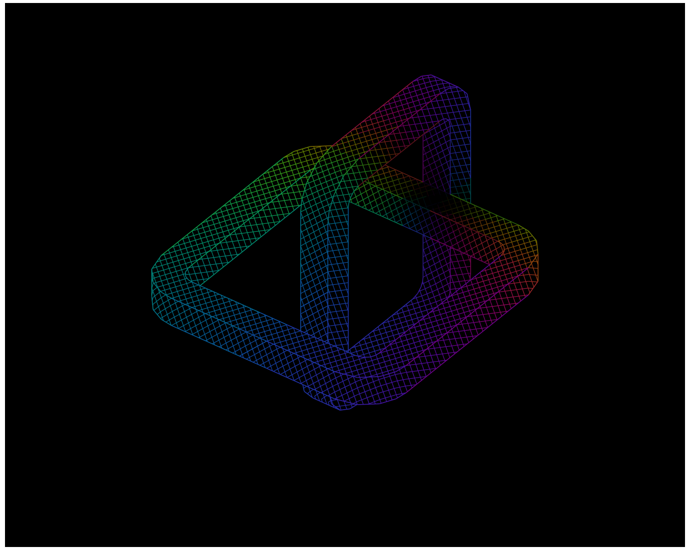
Future-receiver condition. A cached image is sufficient only for the receivers and continuations it preserves. A generator, current and retained fibre can describe indefinitely many later configurations without storing all pictures. ETP's substitution/congruence discipline applies to admitted passages; identical current pixels do not establish it. The companion twenty-plate composition atlas retains the prior HNN/Athena architecture and its exact examples.

PLATE 10

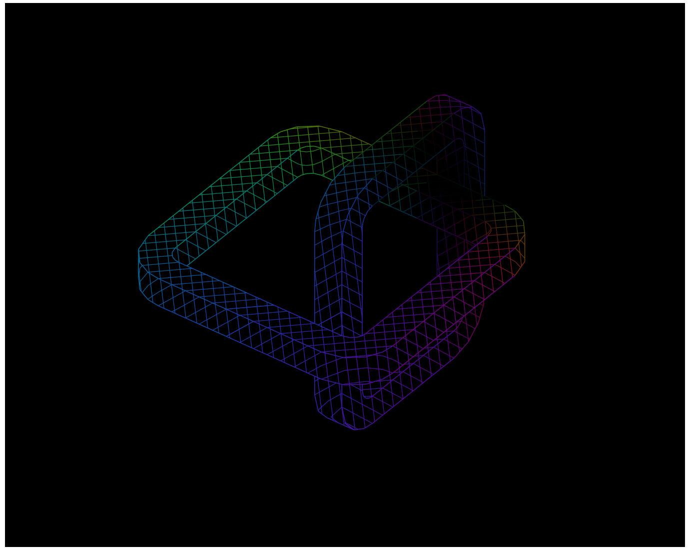
One link changes the motion of the other

Consequential boundary coupling: normal grip, tangential slip, and energy carried into the shared interface.

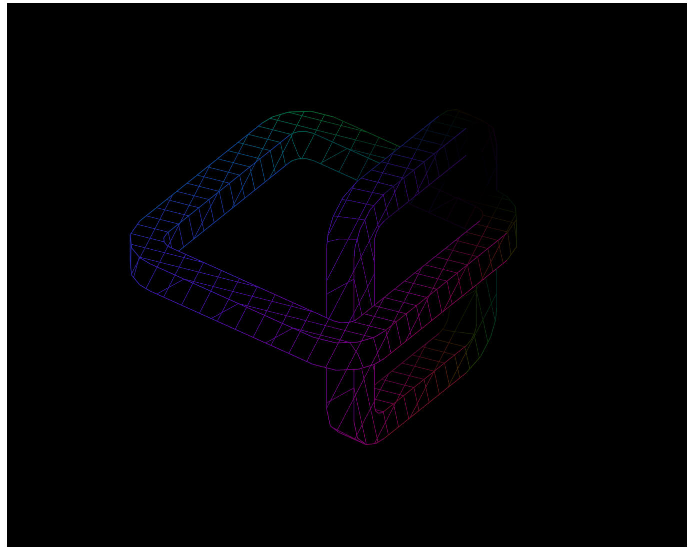
Initial slip · $a = 1$



Coupled motion · $a = \frac{1}{2}$



Reduced slip · $a = \frac{1}{8}$



$$a = e^{-2t}, \quad s = e_y + e_z, \quad v_A = as, \quad v_B = -as, \quad c_A = \frac{1-a}{2}s, \quad c_B = -c_A$$

$$f_A = -2as = -f_B, \quad E_{\text{kin}} = 2a^2, \quad P_{\text{friction}} = 8a^2, \quad Q = 2\left(1 - a^2\right)$$

A real shared patch constrains the movement.

$$g \geq 0, \quad N \geq 0, \quad gN = 0, \quad g = 0, \quad N = 1$$

Exact guided-body model. Two unit-mass rounded links have disjoint interiors and a shared square of area $\frac{1}{4}$. Opposed normal preloads are balanced by contact reactions. Guides hold their rotations; tangential motion follows unit viscous contact. This is a declared contact law, not an unconstrained elastic-knot solver.

Grip and turn are measured together.

$$A_R = \frac{r \cdot F}{\mathrm{E}_0} + \mathrm{i}\frac{(r \times F) \cdot n_R}{\mathrm{E}_0}$$

The body hatches read this work/turn pair with $E_0 = 2$, $F_A = -Nn + f_A$ and $F_B = -F_A$. Color follows the same primary law. The links retain linking number -1; no surface passes through another. The buried contact is read separately on the next plate by an interface receiver.

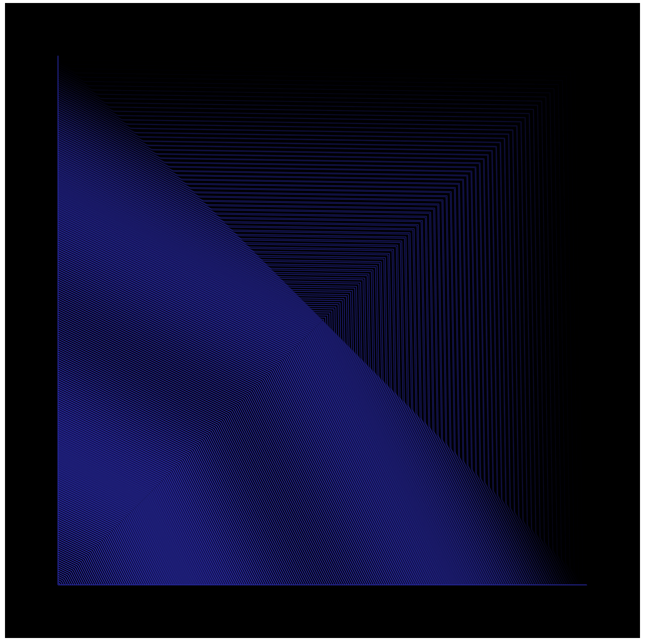
Recovered meaning. The Turn tablet defines friction as the consequential coupling itself, with grip and slip as two faces. The positive heat return here belongs to this particular viscous constitutive law. At the displayed phases, $t = -\frac{\log(a)}{2}$; positions, forces and energy are exact ratios, rather than an interpolated animation.

PLATE 11

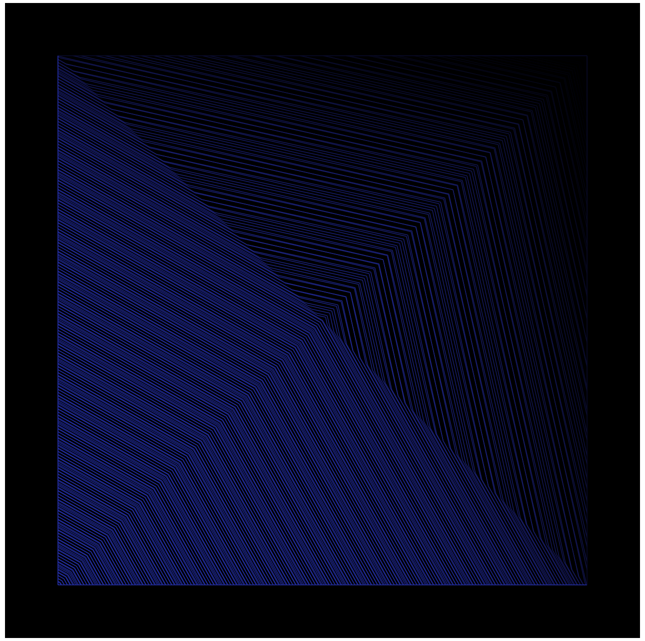
Friction drives a complex interface current

Momentum transfer and heat propagate through the actual boundary adjacency of the contact patch.

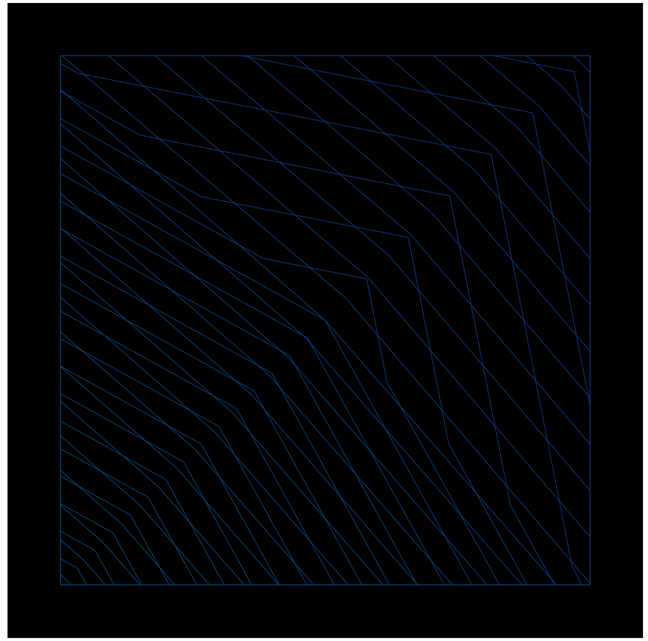
Early contact · $a = \frac{255}{256}$



Propagation · $a = \frac{15}{16}$



Distributed return · $a = \frac{1}{2}$



$$L = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}, \quad \ell = 2r = \frac{1}{2}, \quad \frac{D}{\ell^2} = 4$$

$$\dot{\theta} = -4L\theta + 8a^2 e_0, \quad \dot{\Psi} = -4L\Psi + 2(1+\mathrm{i})ae_0, \quad \Psi \in \mathbb{C}^4$$

$$\sum_i \theta_i = 2(1-a^2), \quad \sum_i \Psi_i = (1+\mathrm{i})(1-a), \quad E_{\mathrm{kin}} + \sum_i \theta_i = 2$$

Exact finite propagation. Four boundary nodes have unit thermal capacity and unit diffusivity at their actual spacing. Frictional power enters one declared port; the operator transports it along contact edges. Ψ is the complex diffusing response to transmitted tangential force, with its sum equal to the delivered impulse. It is a history receiver, not an additional uncounted mechanical energy store.

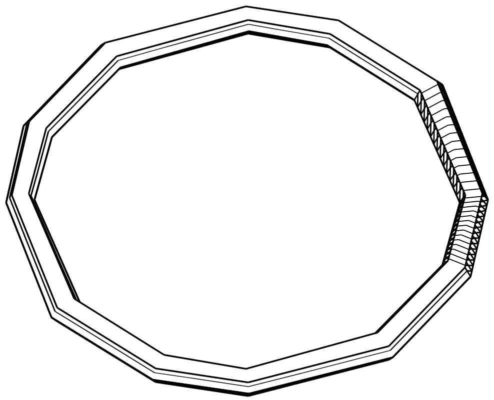
Declared interface image. Four fan triangles interpolate the boundary readings; the center is their mean, not a fifth dynamical state. Ink reads normalized impulse rate together with stored heat. The fixed receiver aperture is the per-node quarter-area $\frac{1}{16}$, and the field-level spacing is $\frac{1}{256}$. This internal receiver has access to the contact patch; the outer views on plate 10 do not pretend to see through opaque bodies. The finite interface law accompanies the C^3 bulk constructions above without claiming a complete continuum fluid/contact solver.

PLATE 12

The knot is retained while its measured face changes

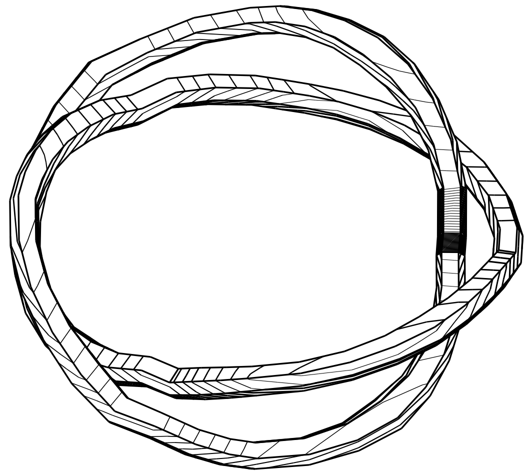
Prime knots and an unknot are explicit closed polygonal carriers, with triangular transverse faces.

Unknot · 0_1



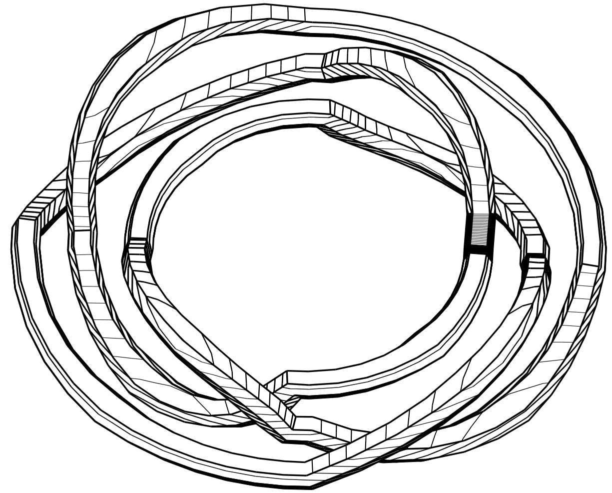
$$\Delta(t) = 1$$

Trefoil · 3_1



$$\Delta(t) = t^{-1} - 1 + t$$

Figure-eight · 4_1



$$\Delta(t) = -t^{-1} + 3 - t$$

$$\text{convex spanning disk},\quad \text{closure}(\sigma_1^3),\quad \text{closure}\Big(\big(\sigma_1\sigma_2^{-1}\big)^2\Big)$$

A crossing is a measured difference in depth. **Established-bounded.** Rational annular braid cells return exactly 0, 3 and 4 projected crossings. The generator checks every nonincident segment pair, the angular cell and the actual over/under sign. The prime labels and Alexander polynomials are standard identifications of these certified diagrams; they are not inferred from hue.

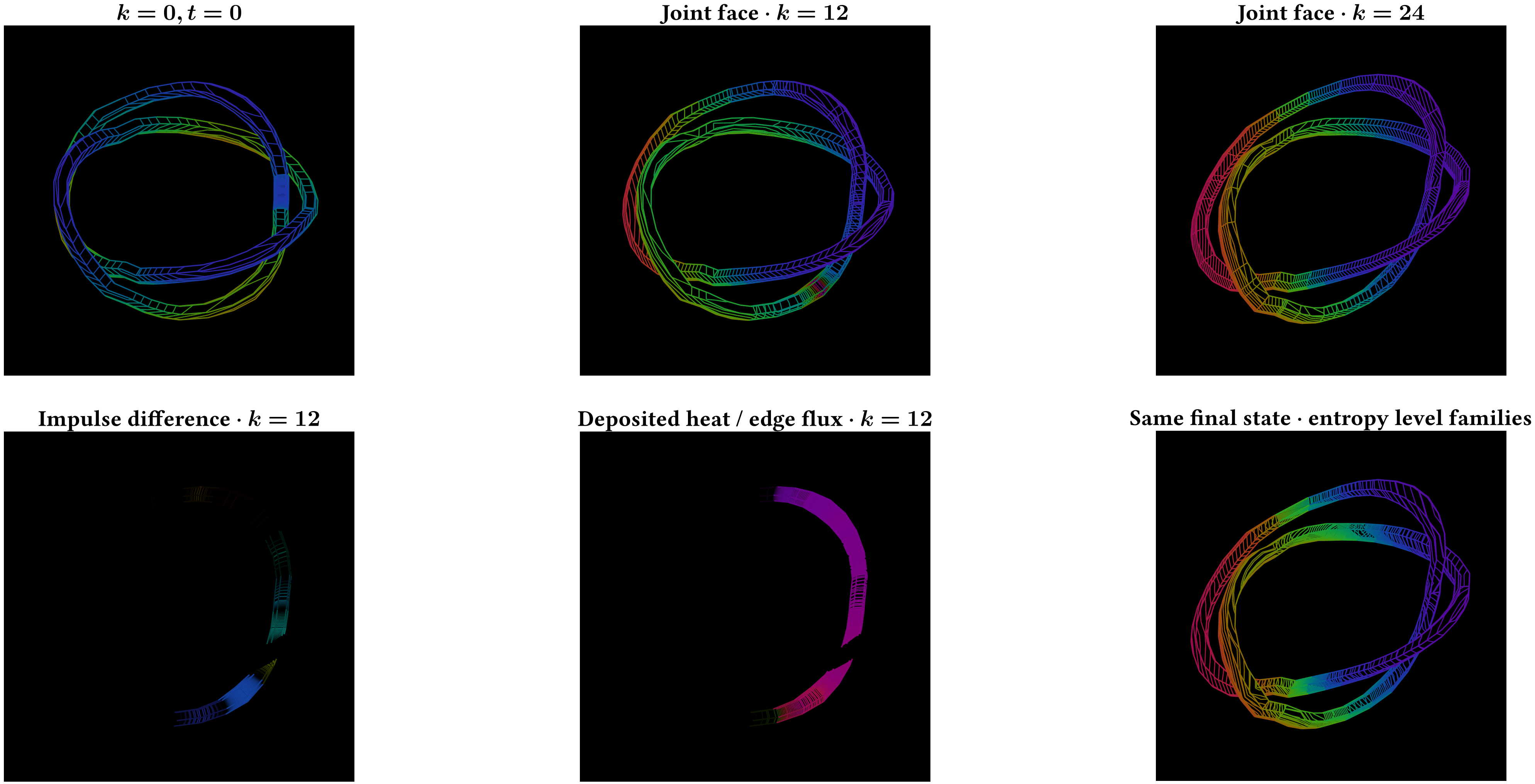
Thickness retains the actual joints. **Established-bounded.** Rational transverse hexagons form a closed oriented triangular tube. All triangle pairs are checked for intersections beyond their shared combinatorial face, including coplanar joints. Each component has $\chi = 0$. The affine fluid map moves entire faces exactly; it cannot undo the knot.

The current Lean torus-slope owner proves coprime embedded circles and the quandle owner supplies local transport laws. Ambient knot classification is carried here by the explicit standard braid diagram and exterior exact checks. Source: Knot Atlas.

PLATE 13

An impulse travels while the trefoil is perturbed

One causal source, two counterpropagating complex ports, a moving body, and an explicitly retained heat return.



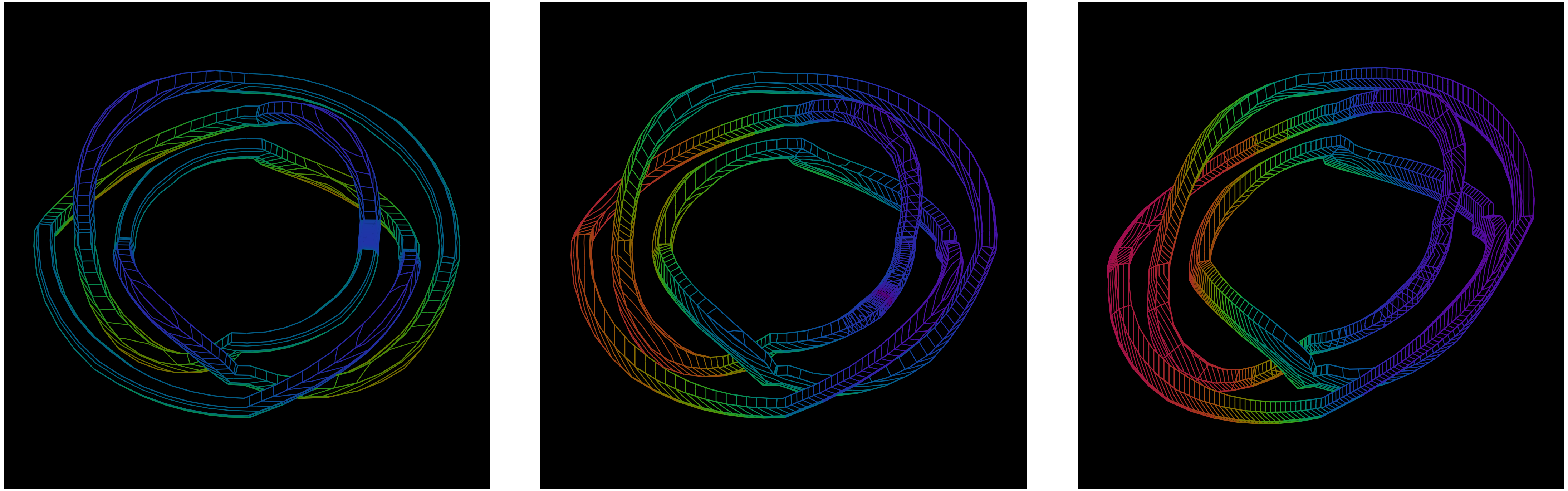
Exact source. Scatter $R = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}$; attenuate by $d = \frac{15}{16}$; move one incident edge per step. Initial ports are $(1, \mathrm{i})$ at one station, zero elsewhere. Heat receives $(1 - d^2)P_i$ and diffuses through self/neighbor weights $\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$. Every step checks total wave energy plus heat $= 2$. Here $t = \frac{k}{24}$. The lower receivers expose the source-relative impulse, heat/edge flux in units $\frac{2}{N}$, and final cross-entropy. Black omission does not erase the retained carrier.

PLATE 14

The figure-eight keeps its crossing law through the flow

A different prime carrier, the same constitutive wave law and the same receiver; topology does not supply a new palette.

$\boldsymbol{k = 0, t = 0}$ $\boldsymbol{k = 12, t = \frac{1}{2}}$ $\boldsymbol{k = 24, t = 1}$



$$F_t(x, y, z) = \left(x + \frac{tz}{2}, y + \frac{t^2 x}{4} + \frac{t^3 z}{8}, z\right), \quad \det DF_t = 1$$

$$U_t = \left(\frac{z}{2}, t\frac{x}{2} + t^2\frac{z}{8}, 0\right), \quad f_t = \left(0, \frac{x}{2} + t\frac{z}{2}, 0\right), \quad P = 0, \quad \operatorname{curl} U_t = \left(-\frac{t^2}{8}, \frac{1}{2}, \frac{t}{2}\right)$$

$$\partial_t U + (U \cdot \nabla)U = \nu\Delta U - \nabla P + f, \quad \nabla \cdot U = 0, \quad \Delta U = 0$$

Proved-derived · exact polynomial check. These identities hold holomorphically on $\mathbb{C}^3$. The displayed mesh lies in the invariant real slice. F_t is spatially affine and invertible, so it transports the actual triangular faces and preserves the knot for all real t. The external source f is displayed; this is a forced material-flow realization.

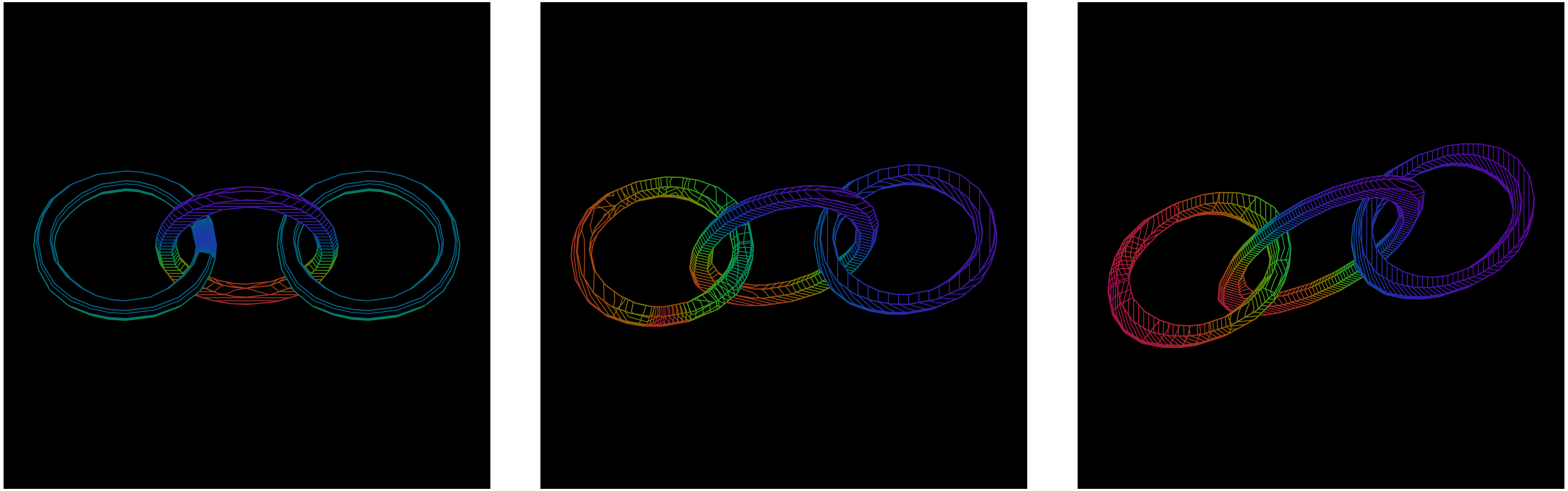
Declared joint receiver. The complex face reads $U + i\ell \operatorname{curl} U + (\Psi^+ + \Psi^-)DF_t v_i$, with material tangent v_i and lane-length chart $\ell = 1$. Velocity, rotation and internal propagation remain separately recoverable. The same coherent primary law gives the colors, with aperture $\frac{1}{16}$.

PLATE 15

A chain restricts motion before a contact is excited

Three unknotted components retain their linking under the same bulk perturbation; an impulse starts on the first component.

$k = 0, t = 0$ $k = 12, t = \frac{1}{2}$ $k = 24, t = 1$



adjacent pair: Hopf link, end pair: separated, $F_t^{-1} F_t = I$

Linking constrains admissible relative continuation. **Established-bounded.** Three planar polygonal cores have adjacent spanning-disk intersections. Their closed triangular tubes are disjoint. The common volume-preserving flow carries the chain without crossing. This fixes topology while allowing its receiver shape and orientation to change.

Contact supplies the actual exchange law. **Definition.** Internal wave incidence stays on each separated component; an impulse does not jump across a projected crossing. Once an admitted contact is reached, its grip/slip and work ports couple the bodies. Plates 10–11 already derive that exchange, frictional heat and diffusion from a real common patch.

$$Q_i^{k+1} = \frac{Q_{i-1}^k + 2Q_i^k + Q_{i+1}^k}{4} + (1 - d^2) P_i^k, \qquad \sum_{i(P_i^k + Q_i^k)} = 2$$

Proved-derived · exact finite passage. A local wave impulse, dissipated heat and receiver cross-entropy have distinct returns. Entropy hatching measures the emitted channel distribution; it does not govern topology or replace the signed wave current. The displayed chain is carried by the prescribed bulk source, not a solved free fluid–elastic-body feedback system.

PLATE 16

Stress and current cross the same material boundary

Fluid traction, electromagnetic work and capacitive storage meet through their actual ports.

$$\begin{tikzcd}
\begin{array}{c} T_{\text{fluid}} \\ \text{momentum, pressure, shear} \end{array} \arrow[r, leftrightarrow, "Tn,\quad t\cdot u"] & \begin{array}{c} \Sigma \\ \text{oriented contact / charge surface} \end{array} \arrow[r, leftrightarrow, "\Delta V, I"] \arrow[d, "P_{\text{port}}"'] & \begin{array}{c} F_{\mu\nu}, j^{\mu} \\ \text{electric and magnetic current} \end{array} \\
 & \begin{array}{c} E_{\text{stored}} + Q \\ \text{returned energy and heat} \end{array} &
\end{tikzcd}$$

The fluid return retains rotational momentum.

$$\rho D_t u = -\nabla p + \nabla\cdot\tau + j\times B + f,\quad \tau = 2\mu D,\quad D = \frac{\nabla u + (\nabla u)^T}{2}$$

$$\partial_t u = \nu\Delta u + u\times \operatorname{curl} u - \nabla\left(\frac{p}{\rho} + \frac{|u|^2}{2}\right) + \frac{f}{\rho}$$

Conditional physical chart. Incompressible constant-density Newtonian fluid; the second line uses force-inclusive f. Viscous heat is $2\mu D : D$. Maxwell stress adds both traction and energy flow; a surface normal alone supplies neither.

The same boundary can store an electrical difference.

$$Q_S = \int_S D_{\text{EM}}\cdot n,\quad I_S = \int_S j\cdot n,\quad \Delta Q = C\Delta V$$

$$j = \sigma(E + u\times B),\quad P_{\text{Ohm}} = \int_V \frac{|j|^2}{\sigma}$$

Conditional circuit reduction. The closed boundary reads enclosed free charge; the charge/voltage comparison also needs the medium and boundary law. Here $\sigma > 0$ is scalar. A changing magnetic flux requires the full path emf. Lightning adds ionization, conductivity evolution and moving leader incidence. The existing leader construction supplies that incidence question.

$$E = \frac{1}{2}(Cv^2 + L^{-1}\varphi^2),\quad e = \nabla E,\quad \dot{x} = s - Ge,\quad \dot{E} = e\cdot s - e\cdot Ge,\quad \dot{Q} = e\cdot Ge,\quad \frac{d}{dt}(E+Q) = e\cdot s$$

Proved-derived · Lean. `PortEnergyHeat` derives the last balance from coordinate derivatives and the storage law. $x = (v, \varphi)$ has conjugate $e = (Cv, L^{-1}\varphi)$; the supplied source and conductance coefficients carry the corresponding units. Contact drag uses the same work/return pattern with its own constitutive law.

A receiver reads energy, momentum and tension together

Einstein's source retains energy and stress; the observer changes their measured faces.

$$G^{\mu\nu} + \Lambda g^{\mu\nu} = \frac{8\pi G}{c^4} T^{\mu\nu}_{\text{total}}, \quad \nabla_\mu T^{\mu\nu}_{\text{total}} = 0$$

$T_{\text{fluid}} + T_{\text{EM}} + T_{\text{other}}$ retained constitutive sectors ⟷ $g, \nabla,$ curvature field equation + boundary data → u_R, n_R, Σ_R local energy / flux receiver

$$T^{\mu\nu}_{\text{fluid}} = (e+p)u^\mu u^\nu + p g^{\mu\nu} + \pi^{\mu\nu}, \quad g(u,u) = -1 \quad (c = 1, \pi = \text{shear stress})$$

$$J^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}, \quad \partial_\lambda J^{\lambda\mu\nu} = T^{\mu\nu} - T^{\nu\mu} + x^\mu f^\nu - x^\nu f^\mu$$

Proved-derived · Lean. The angular-current identity assumes $\partial_\lambda T^{\lambda\nu} = f^\nu$ and retains antisymmetric stress. Symmetric stress and zero divergence cancel them in the inertial derivative chart. Spin may contribute an additional current; curved spacetime needs the corresponding connection and symmetry/Killing field for a conserved global charge.

A concrete observer change separates dark-sector proposals.

$$T = \operatorname{diag}(e, p_1, p_2, p_3), \quad u_R = \left(\frac{5}{3}, \frac{4}{3}, 0, 0\right), \quad n_R = \left(\frac{4}{3}, \frac{5}{3}, 0, 0\right)$$

$$T(u_R, u_R) = \frac{25e + 16p_1}{9}, \quad T(u_R, n_R) = \frac{20(e + p_1)}{9}$$

Source chart	Energy	Mixed reading
Dust: $p_j = 0$	$\frac{25e}{9}$	$\frac{20e}{9}$
Vacuum: $p_j = -e$	e	0

Tension is a testable stress choice.

$$e + \sum_j p_j = \begin{cases} e & \text{dust} \\ -2e & \text{vacuum} \\ 0 & \text{axial: } p_1 = -e \\ p_2 = p_3 = 0 & \end{cases}$$

Interpretation. Investigate dark contributions through density, anisotropic stress, exchange and receiver fibres. Vacuum negative pressure and clustering matter have different transported readings. The physical identifications owe lensing, expansion and structure-formation returns; one scalar “tension” cannot identify all three.

$$E^2 = c^2|p|^2 + \left(m_0 c^2\right)^2, \quad p = 0, E \geq 0 \to E = m_0 c^2$$

Proved-derived · Lean. The positive rest branch follows from the existing mass-shell owner with $m_0 \geq 0, c > 0$.

Sources: Tong, General Relativity PDG dark matter PDG dark energy. The existing Einstein–fluid interface derives conservation from the field equation, Bianchi identity and metric compatibility.

PLATE 18

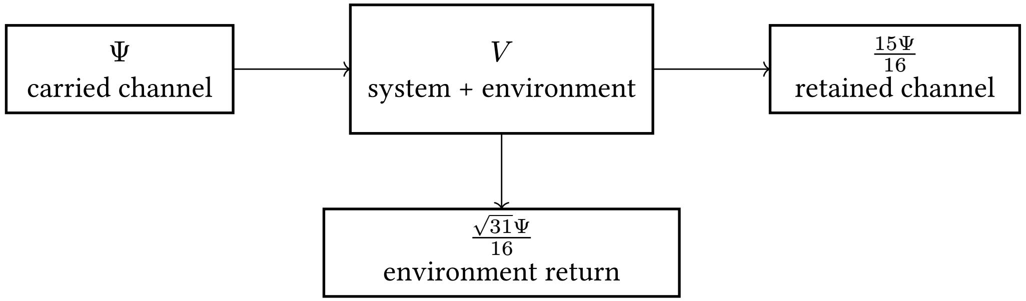
Quantum paths retain amplitudes and interaction rules

A knotted material carrier, a circuit graph and a Feynman graph have different edges; their compositional maps can be explicit.

A local complex wave has a reversible completion.

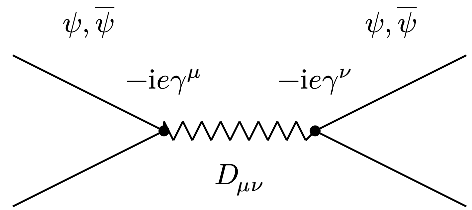
$$R = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}, \quad R^\dagger R = I$$

$$V = \begin{pmatrix} \frac{15}{16} & -\frac{\sqrt{31}}{16} \\ \frac{\sqrt{31}}{16} & \frac{15}{16} \end{pmatrix}, \quad V^\dagger V = I$$



Proved-derived. The same finite scattering map preserves classical complex wave power and quantum amplitude norm. Reading the discarded port as physical heat additionally requires a reservoir and its energy law. $\sqrt{31}$ is derived from $16^2 - 15^2$, not a fitted rendering constant.

A propagator and a vertex make the graph executable.



$$G = G_0 + G_0 \Sigma G, \quad G = \left(G_0^{-1} - \Sigma\right)^{-1}$$

Conditional. The drawn QED exchange has fermion lines, two coupling vertices and a photon propagator. It is a term in an amplitude expansion, not an electron trajectory or a literal knot crossing. The displayed Dyson relation is an exact inverse identity where the inverses exist; its iterated series also needs convergence or a declared formal-series chart.

Sector	Connection / fibre	What the common carrier preserves
Electromagnetic	$U(1)$	Oriented paths, joins and returned face holonomy
Weak / electroweak	$SU(2)_L \times U(1)_Y$	Sector representation and Higgs coupling remain attached
Strong	$SU(3)$	Color representation, vertices and gauge constraints remain attached
Gravity	Frame / Spin connection	Metric, stress-energy source and causal receiver remain attached

Existing formal carrier. `HolonicFourForceSectorCarrier` and gauge-covariance owners retain the common incidence and ordered transport; they do not erase the different actions or quantize gravity. A real Lorentzian stress source from complex fields is derived from a real action, with its conjugate channels retained. The holomorphic C^3 fluid chart does not become that physical source merely by taking its real part.

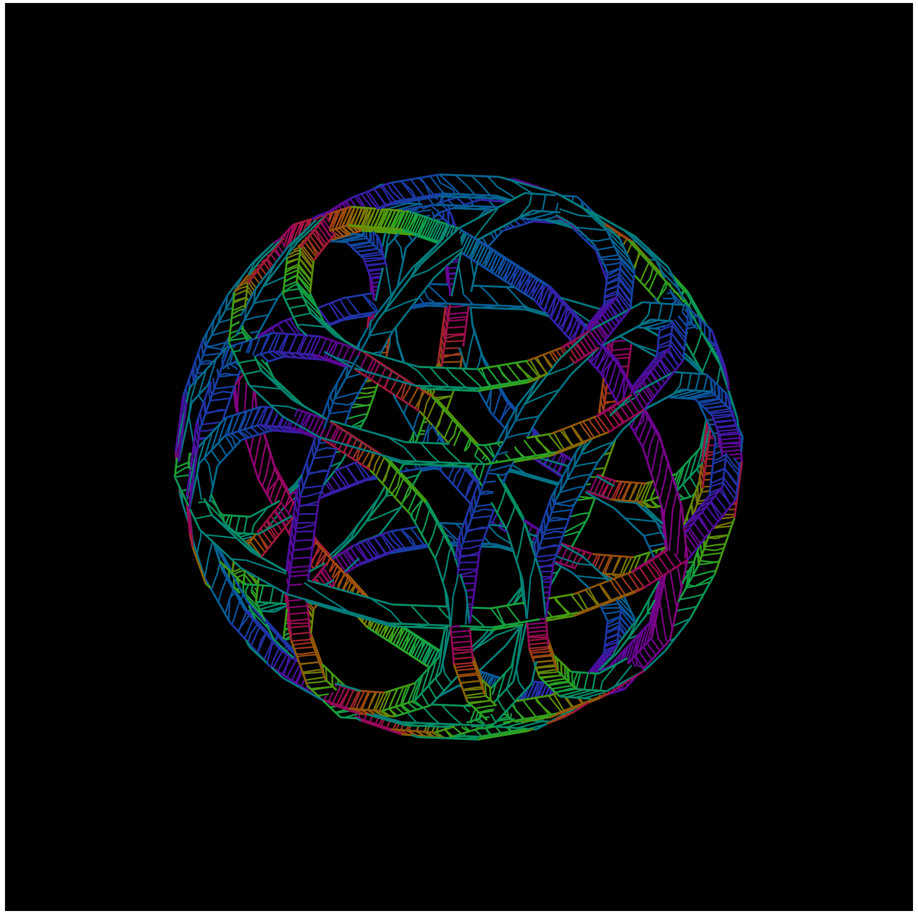
Sources: Tong, Quantum Field Theory, §§1.3, 3.4, 6 Holonics finite quantum transport, Yang–Mills covariance and four-force owners. ETP compares actual substitution/composition laws, beyond a shared graph silhouette.

PLATE 19

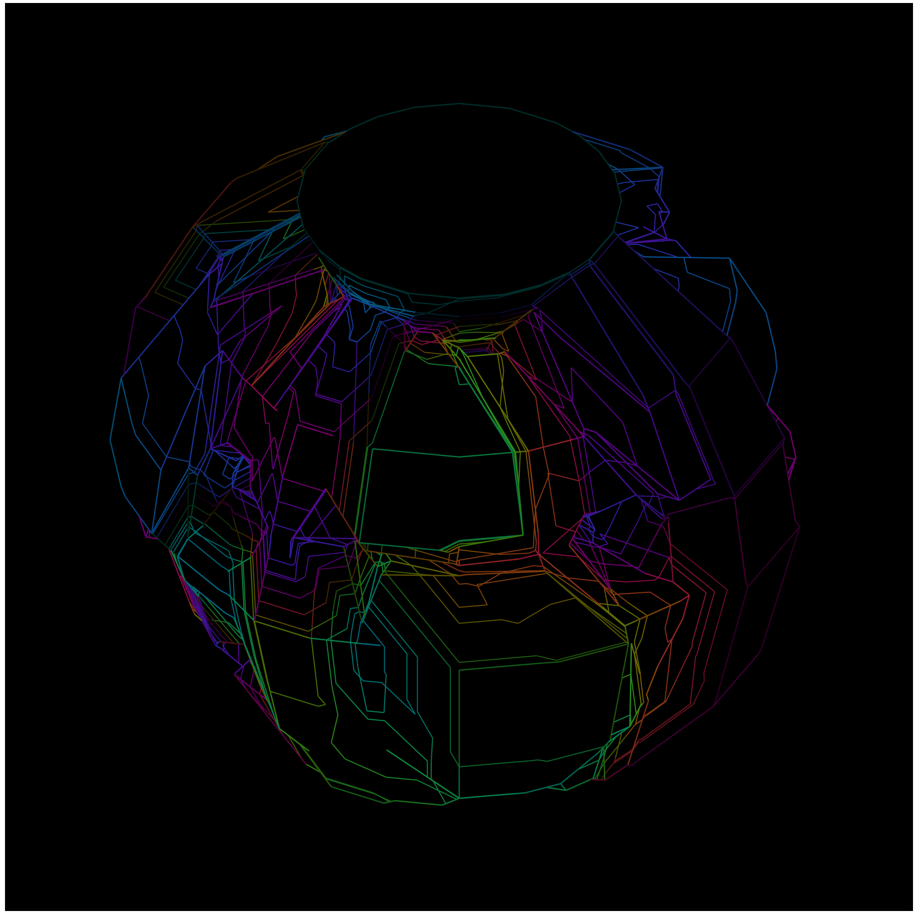
Many circulating fields give a body its visible surface

Fifteen toroidal channels, actual overlap ports, and one coherent receiver; the visible skin is returned by their interaction.

The circulating channels



Their joint intensity horizon



$$A_v = \sum_i \Psi_i K_i(v) g_i(v), \quad A_h = \sum_a \lambda_a A_{v_a}, \quad I_h = \sum_a \lambda_a \left|A_{v_a}\right|^2, \quad \Sigma_R = \left\{x : I_h(x) = \frac{1}{3}\right\}$$

$$I_h - |A_h|^2 = \sum_{a<b} \lambda_a \lambda_b \left|A_{v_a} - A_{v_b}\right|^2 \geq 0, \quad \lambda_a \geq 0, \quad \sum_a \lambda_a = 1$$

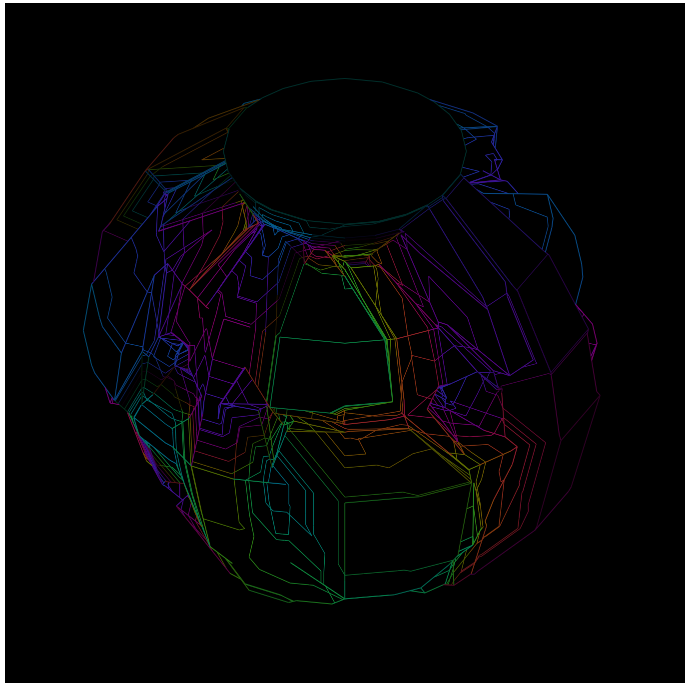
Definition. Three orientation families each carry five rational torus sections. Their full field supports overlap; the thinner lines on the left display the channels. K_i is a compact toroidal kernel, g_i a unit phase plate, and Ψ_i the transported mode. No source incidence is created by a screen crossing.

Exact finite receiver. I_h interpolates the measured nodal intensities on a radially layered tetrahedral atlas. The surface is exact for that finite chart. Intensity averaging retains the displayed phase-variance defect; it is not silently replaced by squaring the averaged amplitude. A dark face need not mean the carrier vanished.

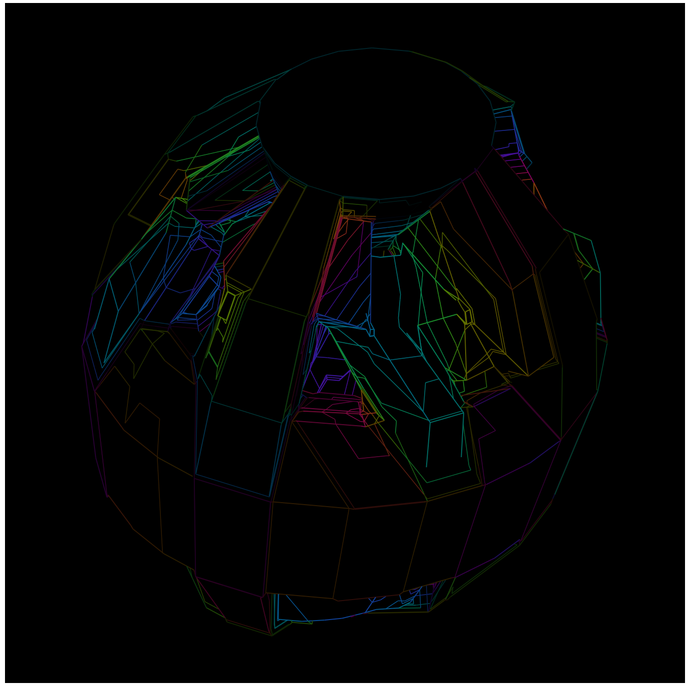
An ecology persists through supplied and returned current

Three illumination ports sustain the woven field; contact transports phase, redistributes amplitude and deposits heat.

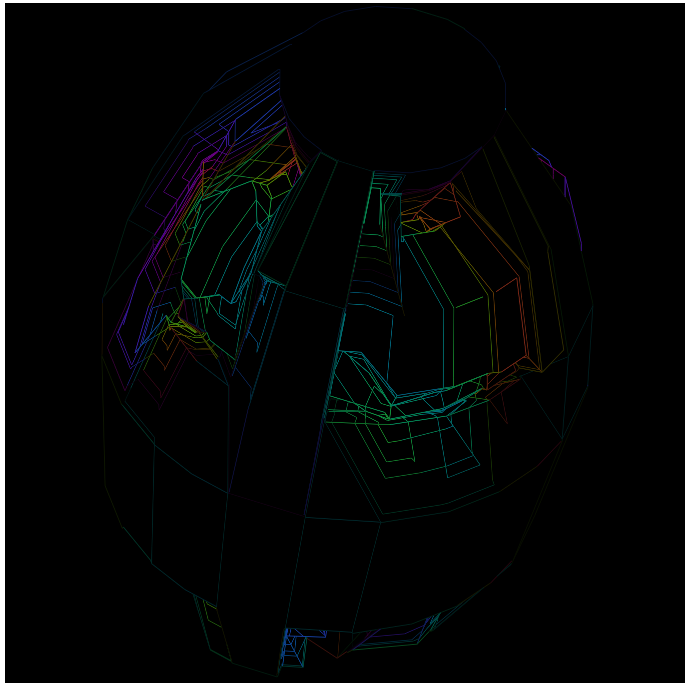
Initial occurrence · $k = 0$



Returned contact · $k = 6$



Continued ecology · $k = 12$



$$\delta_e = y - u_e x, \quad x' = x + \alpha \overline{u_e} \delta_e, \quad y' = y - \alpha \delta_e, \quad |u_e| = 1, \quad \alpha = \frac{1}{4}$$

$$E' - E = -2\alpha(1-\alpha)|\delta_e|^2, \quad Q_e = 2\alpha(1-\alpha)|\delta_e|^2 \geq 0$$

$$\sum_i |\Psi_i^k|^2 + \sum_i H_i^k = E_0 + \sum_{\ell \leq k} W_\ell, \quad W_\ell = |\Psi + s|^2 - |\Psi|^2$$

Proved-derived · Lean and exact source. The passive contact law keeps transported sum and energy plus heat. Each clock admits a matching of independent contacts. Their unit phases are read at common support nodes. The atlas also follows plate 14's fluid map at $\tau = \frac{k}{M}$, with $M = 13$ returned matchings; optical amplitude is carried as a scalar.

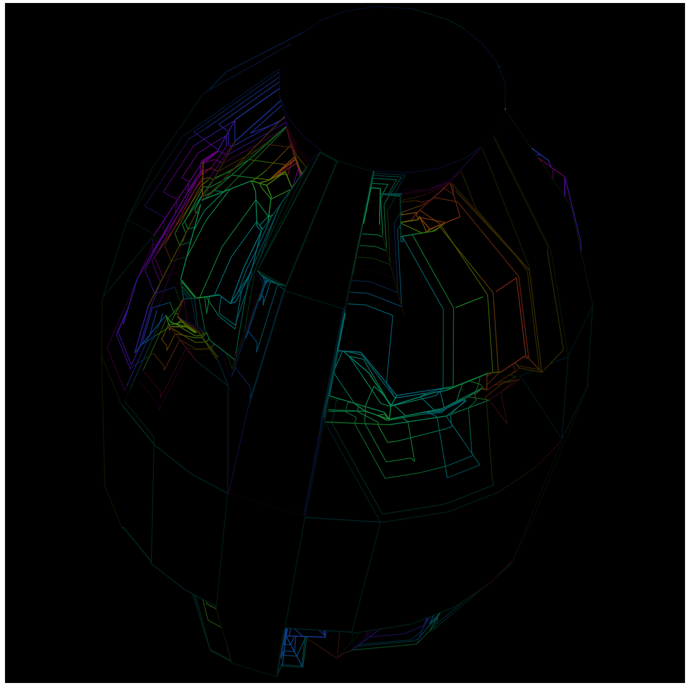
Declared drive. One input port per orientation family supplies a coherent unit amplitude at each clock. The initial three-family pulse enters one channel. The signed source-work cross term is retained; the energy is port storage, not a presumed integral of the image. In the closed-pulse comparison, wave energy becomes heat and the optical horizon can fall below threshold while the channels remain.

PLATE 21

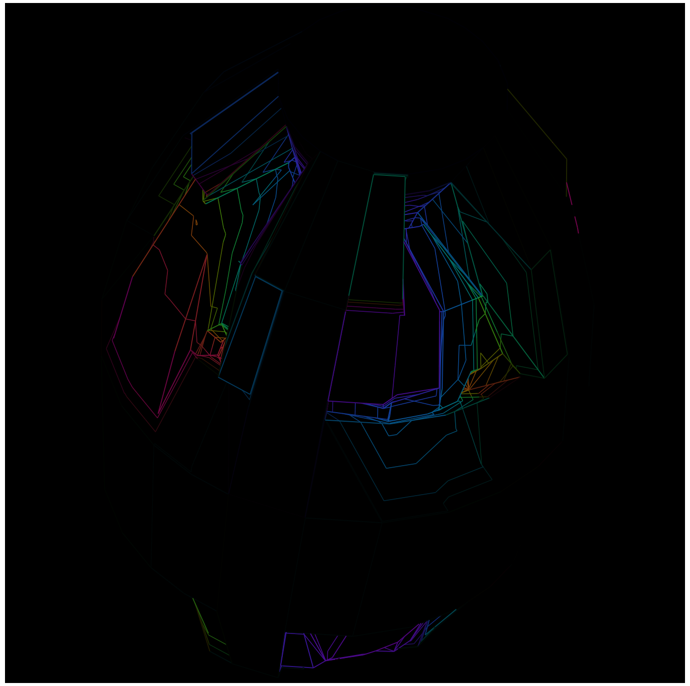
Changing contact and changing the receiver have different returns

The same incoming source and threshold; counterfactual contacts expose what makes the collective face.

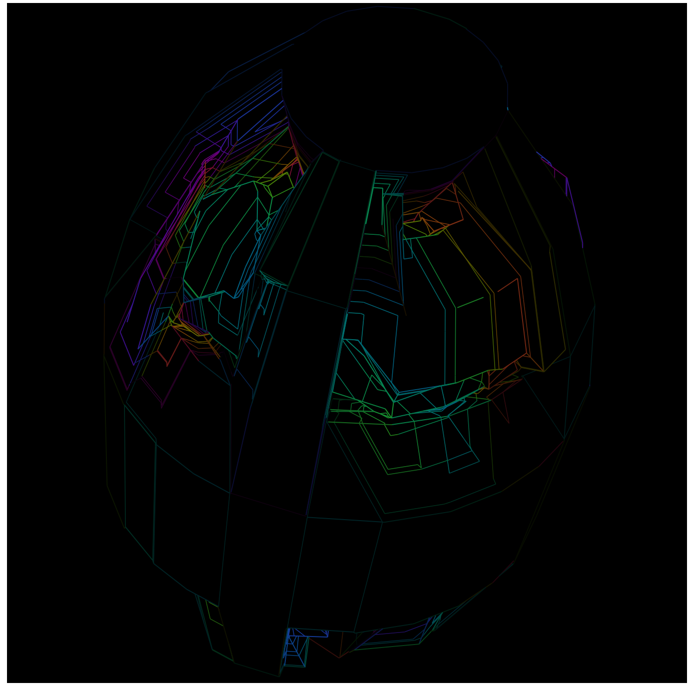
Admitted contact · $k = 12$



Contact disabled · same drive



One reversing seam · same drive



$$u_{ij} = \varepsilon_{ij} g_i(w)\overline{g_j(w)}, \quad \prod_{\gamma_a} u_e = -1, \quad \prod_{\gamma_b} u_e = +1$$

$$\tilde{x} = gx, \quad \tilde{y} = hy, \quad \tilde{u} = hu\bar{g}, \quad \mathrm{contact}(\tilde{x}, \tilde{y}; \tilde{u}) = (gx', hy')$$

Proved-derived · Lean. A coordinated receiver reorientation transports the same contact. Altering one seam while retaining the other ports changes the closed return instead. On the selected common-site spine, the phase-plate factors telescope; the declared sheet sign remains.

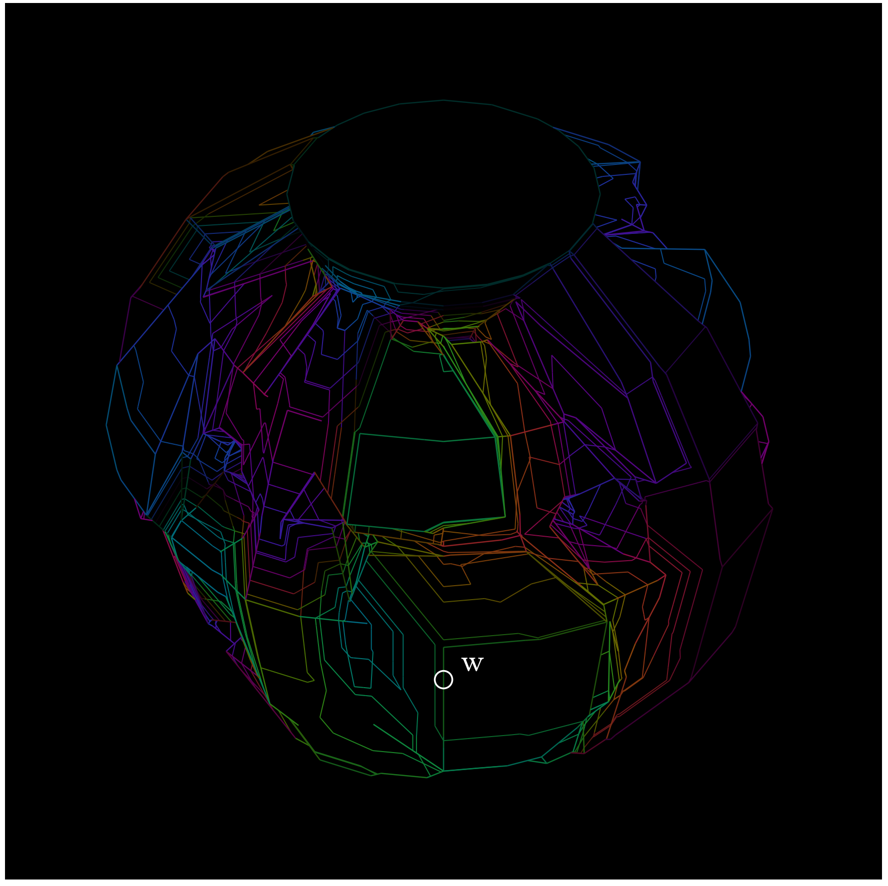
Exact separating instance. A parallel section on $y = x$, $z = y$, $x = -z$ must be zero. The all-positive cycle admits every constant section. Driven current can still traverse the reversing passage; its incompatible local differences and heat remain visible. The source field, not a knot label, determines the image.

PLATE 22

The intersection contains an orientation-reversing passage

A Möbius-shorts field is embedded inside four actual torus supports; its central disk follows the measured local plane.

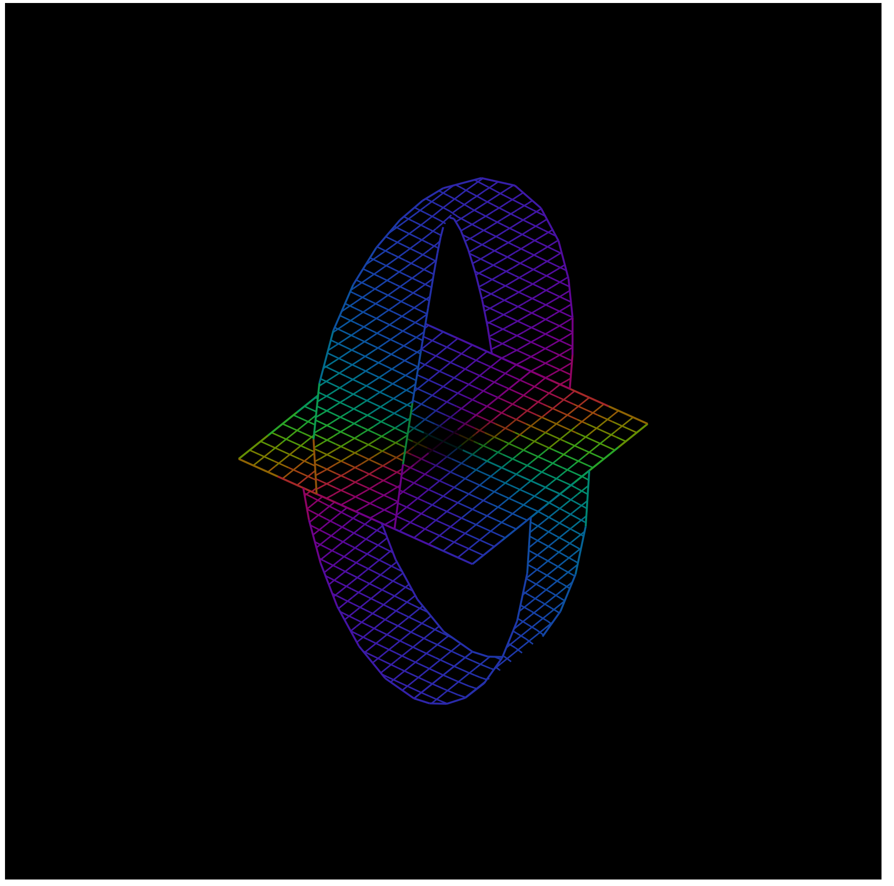
The containing field · w locates the shared source



$$X(x,y,z) = w + \varepsilon(xb_1 + yb_2 + zn), \quad b_1 \parallel \nabla F_i \times \nabla F_j, \quad b_2 \parallel n \times b_1, \quad n \parallel \nabla F_i$$

Established-bounded. Strict rational interval bounds place the whole mapped mesh inside all four toroidal supports. Its source incidence returns $\chi = -1$, one boundary and two loop signs $(-1, +1)$. The affine frame has positive determinant. The disk is tangent to the first field level at the contact and retains the shared tangent direction.

The supported passage · magnified receiver

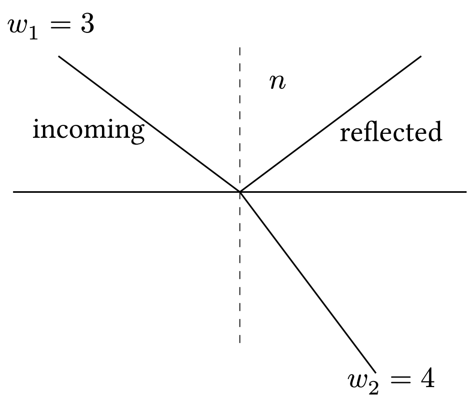


A derived aperture. The physical scale $\varepsilon = \frac{3}{512}$ is the first admitted dyadic scale from the declared search and its whole-box certificate. It is a conservative fitting result, not a universal thickness law. The locator marks the buried source; the inset has its own local receiver. Magnification and full embedding are retained. The fluid horizon and the embedded nonorientable passage remain separate geometric objects.

A local plane carries a law within its measured aperture

Optical tangential transport, a nearly constant gravity reading and a cosmological parameter each retain their own remainder.

Snell transport resolves the normal return.



$$3\sin(\theta_1) = 4\sin(\theta_2) = \frac{12}{5}, \quad (\cos(\theta_1), \cos(\theta_2)) = \left(\frac{3}{5}, \frac{4}{5}\right)$$

$$N_1 = \frac{9}{5}, \quad N_2 = \frac{16}{5}, \quad N_2 - N_1 = \frac{7}{5}$$

$$r = \frac{N_1 - N_2}{N_1 + N_2} = -\frac{7}{25}, \quad t = \frac{2\sqrt{N_1 N_2}}{N_1 + N_2} = \frac{24}{25}, \quad r^2 + t^2 = 1$$

Conditional optical chart. Isotropic lossless TE interface with equal magnetic permeability; t is power-normalized. The Snell owner preserves the tangential receiver while retaining the normal fibre. A reflected phase sign alone does not prove a Möbius surface; the preceding passage also carries its actual gluing incidence.

Near a surface, variation can be small and bounded.

$$\frac{g(R+h)}{g(R)} = \frac{1}{\left(1 + \frac{h}{R}\right)^2}, \quad s = \frac{h}{R} \geq 0$$

$$\frac{g(R) - g(R+h)}{g(R)} = \frac{s(2+s)}{(1+s)^2} \leq 2s$$

Proved-derived comparison. In the spherical Newtonian exterior, “constant surface gravity” is a local aperture approximation with this exact relative defect. It does not make g constant away from the source. The woven contact plane likewise retains its polynomial Taylor remainder over its finite neighborhood.

A constant Λ can have changing receiver coordinates.

$$f(r) = 1 - \frac{2GM}{c^2 r} - \frac{\Lambda r^2}{3}, \quad g_{\text{weak}} = \frac{GM}{r^2} - \frac{\Lambda c^2 r}{3}$$

$$\Omega_\Lambda(z) = \frac{\Lambda c^2}{3H(z)^2}$$

Conditional physical chart. The first line is the static spherical Schwarzschild–de Sitter metric factor and its weak-field inward-positive radial acceleration. A constant geometric Λ does not imply constant Ω_Λ when the expansion chart changes. Our cosmological inference owner retains those plural source/receiver fibres and dimensions.

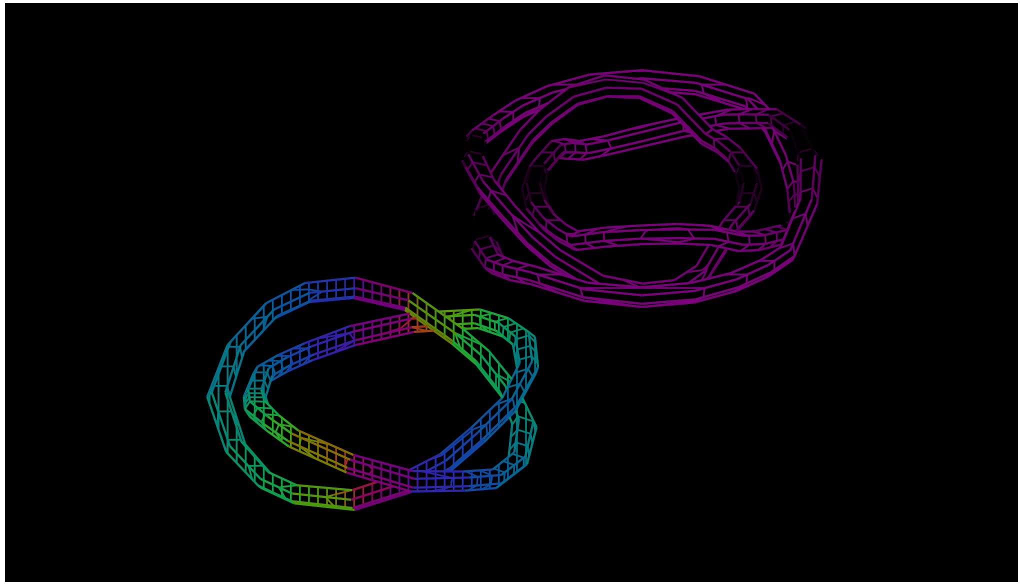
Sources: Tong, GR the August 25 cosmological receiver record; `HolonicSnellInteraction` and the paired phase receiver.

PLATE 24

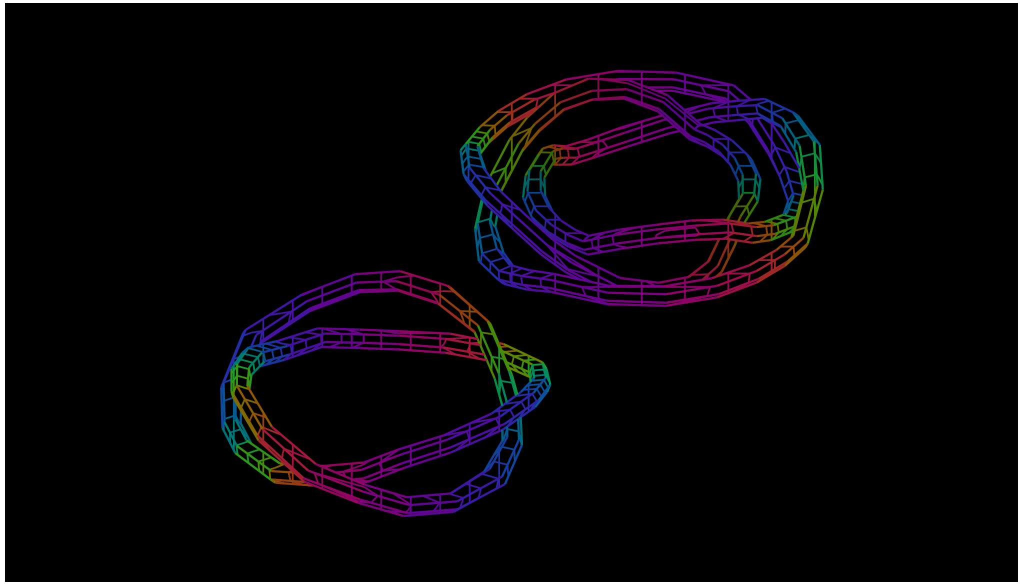
A remote fold responds through its constitutive contact

Two contorted domains form a small abstract allosteric machine: one is driven, while the other moves through the shared response.

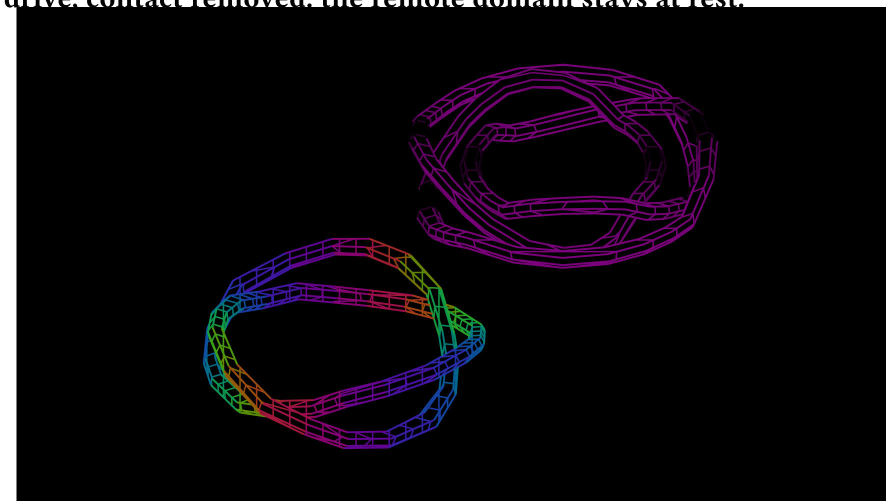
Initial domains · $q_1 = q_2 = 0$



After four returned steps · contact admitted



Same drive, contact removed; the remote domain stays at rest.



Established-bounded. The image reads conformation and generalized force on each actual rotated domain. q parametrizes the rational rotation; it is not a literal angle. The contact-off return keeps $q_2 = 0$, while the admitted contact gives $q_2 < 0$.

$$c_i = \frac{1}{2} + q_i^2, \quad E = \frac{c_1^2 + c_2^2}{2} + \frac{(q_1 + q_2)^2}{2} - q_1$$

$$q' = q - \frac{1}{8}\nabla E, \quad H = F\Delta q_1 - \Delta U = E - E' > 0$$

$$K_i^{\text{domain}} = 2(a + 3q_i^2) = k(2q_i)^2 + 2k(a + q_i^2) \quad (k = 1)$$

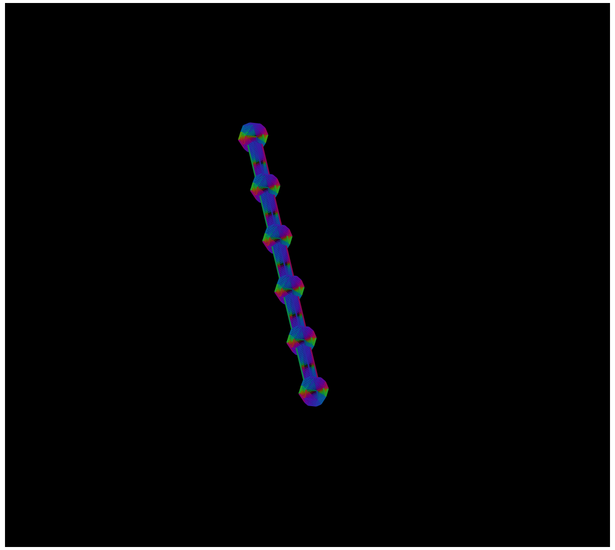
Proved-derived · exact instance. The existing conformation owner separates material stiffness and prestress; the finite source keeps the contact cross term. Every displayed step stays within $|q_i| \le \frac{1}{2}$ and returns positive finite dissipation. The even receiver q^2 can agree while the oriented responses are opposite.

Biological interpretation. This recovers the protein constraint ecology: chains, contact cells, flexible modes, self-stress and an active-site receiver. The construction models remote mechanical consequence and changing constraints. Plates 25–28 now add sequence-conditioned folding and reaction kinetics to this programme.

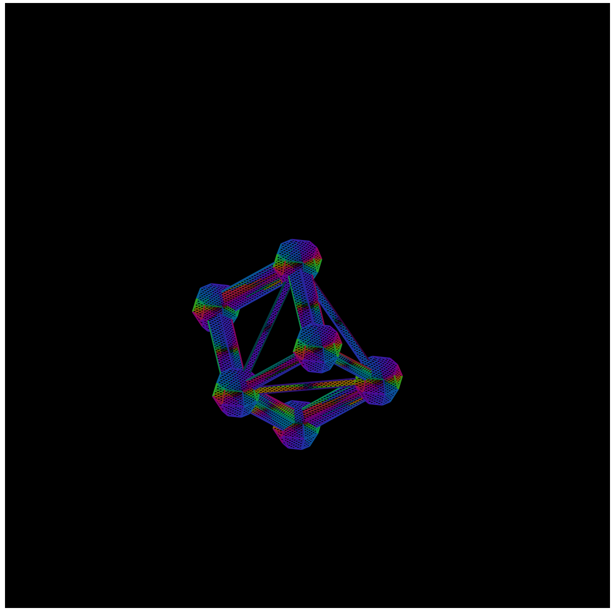
Sequence is a cause of the fold and its subsequent conduct

The same six-site backbone and environment; changing the H/P order changes the energy landscape and kinetic return.

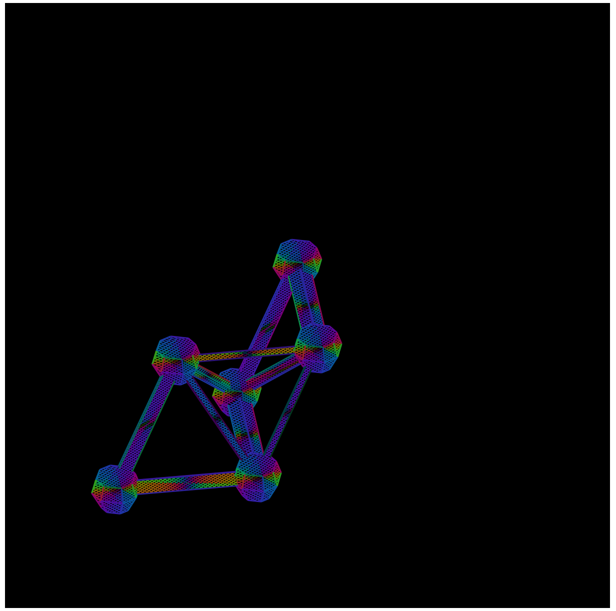
Shared held starting chain



HHPHPH · one minimum-energy conformation



HPHHPH · one minimum-energy conformation



$$q=(r_0,\ldots,r_5),\quad r_0=0,\quad r_1=(1,1,0),\quad \left|r_{i+1}-r_i\right|^2=2,\quad r_i\neq r_j\quad (i\neq j)$$

$$C_s(q)=\sum_{0\le i<j<6,\, j\ge i+2} 1_{s_i=s_j=\mathrm{H}}1_{|r_i-r_j|^2=2},\quad E_s(q)=-\varepsilon C_s(q),\quad \pi_s(q)=\frac{b^{C_s(q)}}{Z_s(b)}$$

Returned family	**HHPHPH**	**HPHHPH**
Anchored conformations	12,711	12,711
Minimum-energy family	16 configurations	16 configurations
Shortest permitted route to that family	16 local moves	20 local moves
Minimum-energy probability at $b=2$	$\frac{512}{33391}$	$\frac{512}{35911}$

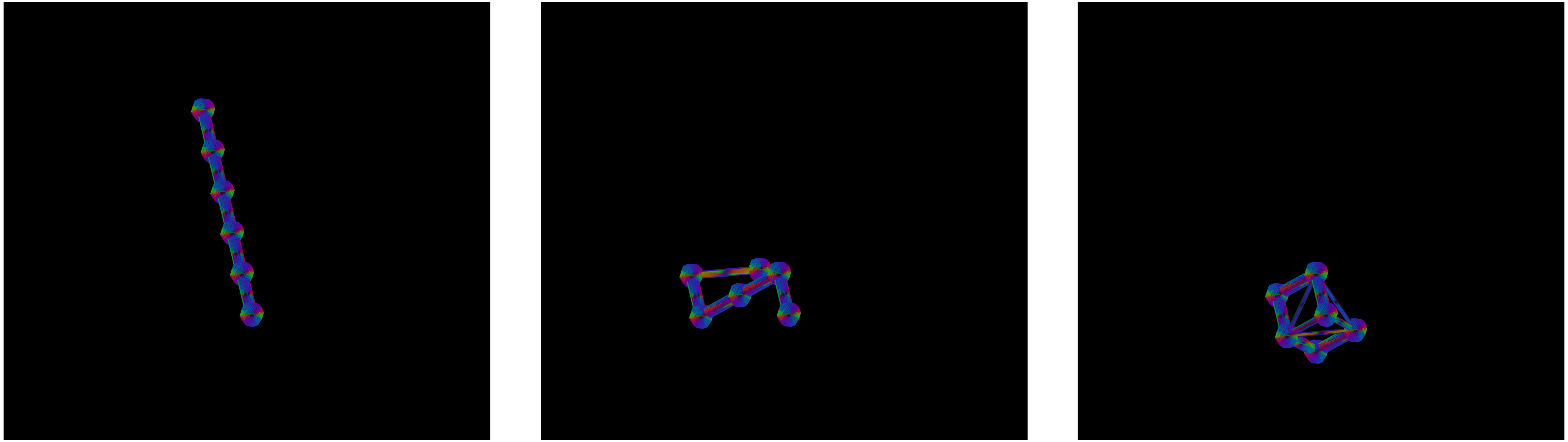
Exact exterior physical chart. FCC nearest-neighbor steps are the twelve permutations/signs of $(1,1,0)$; their cells have triangular faces. H/P supplies a declared hydrophobic/polar contact law. The first bond is held. Both inputs have four H and two P sites. The images show representatives of returned families, with a common receiver for the compact folds; they do not select a unique structure by its label. Color reads actual conditional mobility and contact-energy current through a fixed probe.

PLATE 26

A fold is reached through actual kinetic passages

One shortest permitted source history for HHPHPH; each shown transition belongs to the full retained kinetic graph.

Source · move 0 **A joined intermediate · move 8** **Arrival in the minimum-energy family · move 16**



$$P_s(q,q') = \frac{1}{48}\min\left(1, 2^{C_s(q')-C_s(q)}\right), \quad q' \text{ is a legal local move}$$

$$P_s(q,q) = 1 - \sum_{q'\neq q} P_s(q,q'), \quad \pi_s(q)P_s(q,q') = \pi_s(q')P_s(q',q), \quad p_{k+1} = P_s^T p_k$$

The sequence changes rates through physical contact. **Established-bounded.** The proposal clock has $4 \times 12 = 48$ choices: four mobile sites and twelve FCC hops. A hop retains all bonds and exclusion; rejected proposals stay in the current state. At $\beta\varepsilon = \log 2$, every transition and population is rational. The enumerated graph is connected and has 79,476 directed legal moves.

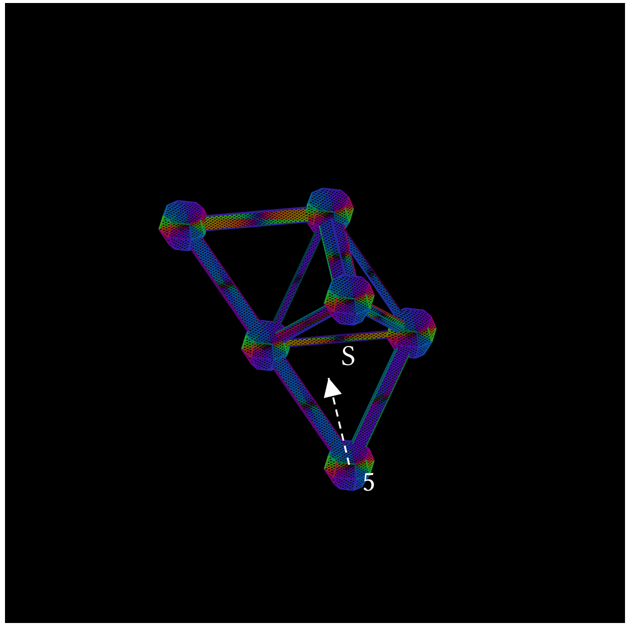
A short permitted history is a source-fibre representative. The displayed path is selected from legal histories by arrival length, with every edge having positive kinetic weight. It is not claimed to be the typical thermal trajectory. The full population, energy degeneracy, initial conditions and clock remain behind the view. No random sampling chooses the shown fold.

Sources: Lau–Dill contact-model precedent Metropolis et al. the current construction uses its explicitly declared three-dimensional FCC chart.

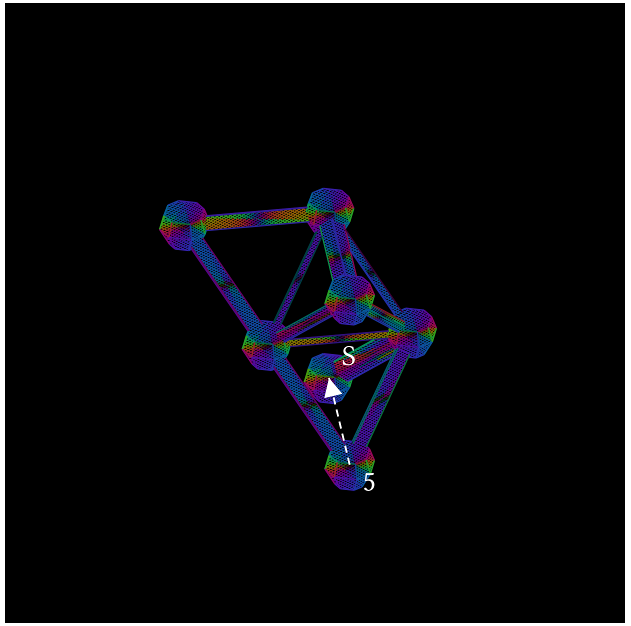
Fold and ligand occupancy change the next available operation

The bound partner occupies a real site; a backbone hop that was permitted now returns to the held state.

Unoccupied binding site · candidate hop has weight $\frac{1}{96}$



Ligand bound · the same hop has weight 0



$$r_4' = r_4 + (1,1,0) = r_5 + d, \quad d = (0,1,-1)$$

$$E + S \underset{k_{\text{off}}}{\overset{k_{\text{on}}a(q)}{\leftrightarrow}} C \xrightarrow{k_{\text{cat}}} E + P$$

$$\dot{x} = \begin{pmatrix} -1 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} k_{\text{on}}a(q)ES \\ k_{\text{off}}C \\ k_{\text{cat}}C \end{pmatrix}, \quad x = (E,S,C,P)$$

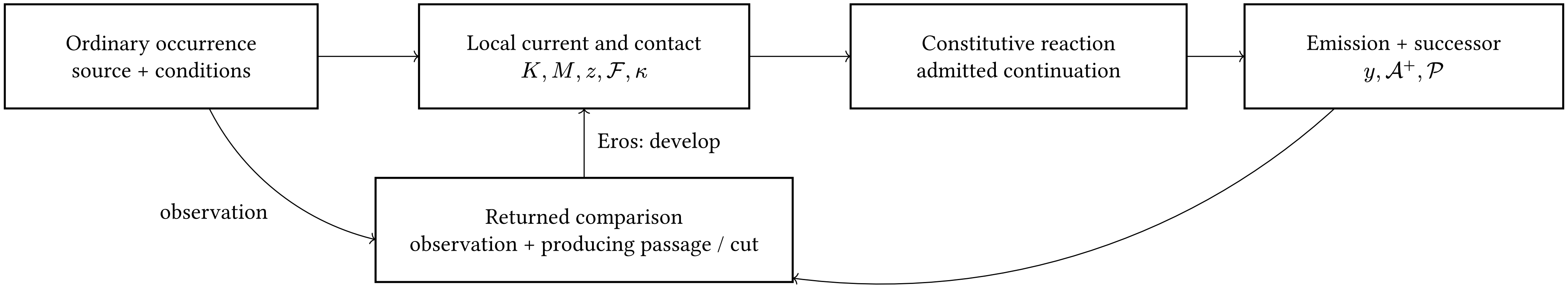
$$\frac{d}{dt}(E+C) = 0, \quad \frac{d}{dt}(S+C+P) = 0$$

Exact mechanochemical source. A terminal ligand binds only in an unoccupied FCC direction. A bound fold move is retained only when the carried ligand also remains collision-free. The model keeps joint (q, free) and (q, bound, d) populations, product count and net substrate drawn. The displayed conservation equations apply to the reaction source; the experiment retains its substrate supply separately. Their exact ledger is bound + product = drawn.

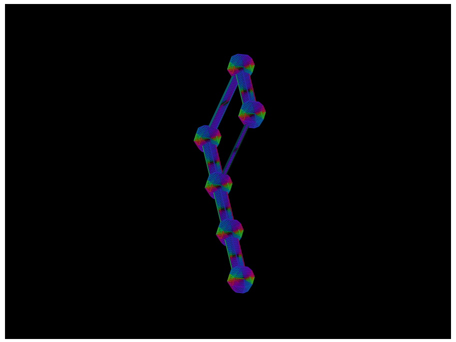
Proved-derived · Lean. `ReactionCurrent` derives the conserved enzyme and substrate moieties from the actual stoichiometric source. The finite kinetic return distinguishes the two supplied sequences: $2^{-26} < \Delta\langle P_{24}\rangle < 2^{-25}$. That small scalar difference accompanies a full retained difference in fold/occupancy history. Product is an expected turnover count, not a probability bounded by one.

A causal thought chain is a changing operative body

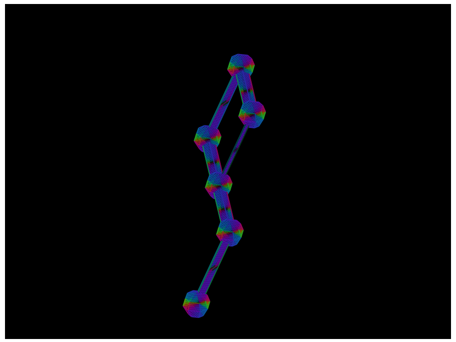
Sequence, local organization and kinetic continuation are foundational HNN questions; the next operation consumes the actual returned state.



One energy face does not determine the next fold.



$C(q_A) = 1$



$C(q_B) = 1$

$$\Pr(C' = 2 | q_A) = 0, \quad \Pr(C' = 2 | q_B) = \frac{1}{48}$$

Exact separator. Including all self loops and rates, the two equal-contact states have different next-bin returns. Thus energy/contact count alone cannot close this kinetic dynamics. Equal fold and free/bound marginals can likewise conceal different binding flux.

The operative chain carries its joining state.

$$X \leftarrow W_f \rightarrow Y, \quad Y \leftarrow W_g \rightarrow Z, \quad W_{g \circ f} = W_f \times_Y W_g$$

$$(\mathcal{A}, o) \rightarrow (y, \mathcal{A}^+, \mathcal{P}), \quad qT_i = U_i q$$

$$z = (p, c), \quad \varphi(z) = (c - p, c, p), \quad \hat{v} = c + M\varphi(z)$$

HNN construction. The shared object is caused source incidence, current, reaction and successor. A molecular codec may supply chemical constraints; text and acoustics supply their own situated sections. A causal thought trace retains the actual joins, conditions and changes that determine later conduct. Verbal output is one receiver of that structure.

$$\mu' = T\mu, \quad \nu' = T\nu + T_{\text{emit}}\mu$$

The kinetic return tracks occupation and labeled output. Unbinding and catalysis can restore the same enzyme state while emitting different outputs.

Concrete handoff. Reuse normal-wave source action and continuation, the condition neighborhood, and the retained producing comparison. Bind exposure/codec associations to the actual source/current and material cut. The existing section-pair development port does not by itself attach an earlier prediction's bounded joint to its returned observation. The revised AC blueprint names that remaining binding; this research does not restart the paused implementation.